

Business Cycles and Police Hires*

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Abstract

We show that the quality of police hires varies over the business cycle. Officers hired when the unemployment rate is high have fewer complaints, disciplines, and are less likely to be fired than officers hired when the unemployment rate is low. Effects are larger for younger workers who have weaker outside options in recessions. We find that the size and quality of the applicant pool increases in high unemployment years—more people take entry exams and a larger fraction pass the exam. Our findings shed light on how outside options affect police hires and speak to policy questions about police recruitment.

Keywords: Police hiring, police quality, public sector labor markets, outside options

JEL Codes: J24, J33, J45, K42

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1 Introduction

The public-sector workforce is the backbone of a well-functioning state. Despite the critical importance of this input in the production of effective government, the best approach to recruitment and selection of public-sector employees remains unclear. While offering financial incentives may attract higher-skilled applicants, it might crowd out applicants with an intrinsic desire to work in government (Bénabou and Tirole, 2003; Delfgaauw and Dur, 2008; Barfort et al., 2019). This tension between extrinsic and intrinsic motivations raises an important question: should government agencies use financial incentives to attract workers?

In this paper, we study the power of financial incentives in the context of police recruitment and selection in the United States. Specifically, we examine how the quality of police hires responds to the availability of outside options in the labor market. We use the unemployment rate at career start as a source of variation in the relative financial attractiveness of policing, as recessions may improve earnings and the likelihood of finding a job in policing relative to outside options (Nagler et al., 2020). Notably, we adopt a broad view of ‘financial incentives’ which encompasses wage offers and performance pay as well as increased hiring in recessions.¹ Under this view, our analysis thus informs how changing the *financial attractiveness* of policing affects new police hires, but does not provide direct guidance on the type of financial incentive agencies should offer.

Studying these issues in the context of policing is both substantively important and policy relevant. First, over 1 million people are employed as police officers in the U.S., making it one of the largest public-sector occupations in the country. State and local governments in the United States spent an estimated \$123 billion on police protection in 2019, with approximately 60% of that total funding police salaries (U.S. Census Bureau, 2020a). Second, although the role of police is to promote public safety, a long history of police violence, racial harassment, and a host of other negative police-civilian interactions have cast doubt on the ability of police to fulfill that role (Pierson et al., 2020; Harrell and Davis, 2020; Ba et al., 2021; Hoekstra and Sloan, 2022). Over the last two decades, public confidence in police has fallen from a peak of 64% in 2004 to 51% in 2021 (Brenan, 2021). Some proponents of police reform have proposed reducing the size of police forces, while others have argued for a focus on hiring higher quality officers. The latter approach, however, is stymied by recent increases in officer quit rates and lulls in recruiting (Shalal and Moore, 2020).

Furthermore, the nature of policing differs in two core ways from other public-sector occupations. First, the intrinsic motivation of police may be associated with negative or

¹For example, we argue that counter-cyclical hiring is one such policy agencies could enact that could induce financially motivated individuals to enter policing during downturns.

positive outcomes. If the intrinsic motivations of police are biased against the interest of suspects (Prendergast, 2007) or driven by a taste for harsh punishment (Dharmapala et al., 2016), then crowding out those applicants might improve police treatment of civilians. On the other hand, if the intrinsic motivations of police are associated with positive outcomes, then financial incentives may crowd out desirable applicants. Second, police training and police culture might transform new recruits in a way that negates gains made on the recruitment margin (Adger et al., 2022).² It is an open question whether it is possible to improve policing via recruitment or worker selection at all, and if so, whether it can be accomplished with financial incentives.

This paper provides the first evidence on these questions by studying how economic conditions, and hence financial incentives, shape the quality of officer hires. We study this in the context of Florida, which has the fourth largest police workforce in the United States, representing nearly 6.5% of nationwide police employment since 1980. We leverage a detailed database of all police employment spells in the state from 1985-2015. These records contain information on officer demographics, employment start date, separation date, and multiple outcomes related to officer quality. We observe all agency-sustained complaints made against an officer, corresponding disciplinary actions, and if they separate from the agency, the reason for separation. We link these records to annual data on economic conditions and assess the relationship between the unemployment rate in an officer’s starting year and that officer’s quality.

We show the quality of new hires does vary in response to economic conditions at career start. Over these three decades in Florida, officers hired when the unemployment rate is higher have subsequently fewer complaints filed against them and fewer disciplinary actions. Officers who enter policing when alternative job opportunities are more scarce are also less likely to be fired. Since a significant share of officers leave their jobs early into their careers, we focus on the officer’s first five years on the force. In our main specification, which controls for agency fixed effects and contemporaneous economic conditions, we find that officers hired when the unemployment rate is one percentage point higher, receive 8% fewer complaints and 13% fewer disciplinary actions within the first five years of their start date. They are also 5% less likely to be fired and 5% less likely to be fired for violating agency policy in those first five years. Furthermore, the effect of initial unemployment across the four outcomes emerges as early as the second year since hire, indicating our results are not driven by early-career attrition.

²It is also possible that police training maintains or amplifies quality differences that exist at the time of recruitment. This *ex ante* uncertainty about how differences at recruitment do or do not persist is a key motivation for our research question.

We show that our results are robust to including agency-by-year fixed effects, thus accounting for the possibility that the nature of policing may vary over time. Although our main specification focuses on the first five years of an officer’s career, our results are also robust to relaxing this restriction and including all years of the employment spell. We also show that our results are robust to including agency-specific linear trends in starting year, controlling for various leads and lags of initial unemployment, removing the contemporaneous unemployment rate as a control, and controlling for officers’ characteristics. We find similar results when using unemployment rates across seven sub-state regions as our measure of economic conditions at career start. Moreover, we find larger impacts for younger officers across the four quality measures — this group may exhibit stronger sorting responses to economic conditions, since firms are less likely to hire young workers during recessions (Forsythe, 2022).

From these results alone, we do not know if the changing quality of hires is due to changes in the pool of applicants or changes in agency hiring decisions. To provide direct evidence on labor supply, we take advantage of additional data on entry exams and on the number of officers hired each year. This allows us to assess the relative importance of labor supply and labor demand in driving our results.³ It is possible that both of these forces are important in this setting. If Florida police agencies are more constrained when unemployment rates are high, they may be more selective in their hiring decisions. On the other hand, workers might find policing more appealing when alternative job opportunities are scarce.

Why might we expect the applicant pool to change in response to economic conditions? First, prior work has documented that the labor supply of teachers and nurses changes across the business cycle (Propper and Van Reenen, 2010; Britton and Propper, 2016; Nagler et al., 2020). Compared to these public sector occupations, policing has relatively low entry costs. In Florida, individuals can enter policing after completing a training program that is typically only 20-30 weeks long. Second, since public sector employment is less pro-cyclical than in the private sector (Kopelman and Rosen, 2016), more high-skilled individuals may apply to become officers in recessions due to higher expected earnings (via increased wages or a higher job-finding rate) relative to other jobs. In our setting, we show that starting salaries in policing in Florida do not vary across the cycle, in contrast to the pro-cyclicality of starting wages in the private sector.

To assess the importance of the labor supply channel, we draw on a dataset of entry-level exams taken by police applicants in Florida from 1996-2015. These exams are taken after

³We also assess the possibility that the nature of policing may change during recessions. We do not find significant changes in the types of agencies that officers joined across the cycle or in their job training participation. Our results are also robust to directly controlling for current and starting year crime rates.

applicants have finished basic recruit training but before they are hired. Not all applicants pass the exam and not all those who pass are hired, making this a good proxy for labor supply. We find that the number of people taking the exam in a year is positively correlated with the unemployment rate. We further show that the fraction of people who pass their initial exam attempt is higher when the unemployment rate is higher. We then confirm that passing the exam on the first attempt is a good proxy for officer quality. First, agencies are significantly more likely to hire officers who pass the exam on their initial attempt. Second, conditional on being hired, officers who pass the exam on their initial attempt have fewer complaints, fewer disciplinary actions, and are less likely to be fired than officers who do not pass on their initial attempt.

Motivated by the finding that passing on the initial attempt is correlated with officer outcomes, we use machine learning to predict agency-sustained complaints and disciplines for each applicant, leveraging the information from the exam, other applicant characteristics and their police academy training records (Chalfin et al., 2016). We first verify that these measures capture a meaningful dimension of quality by showing that our measures of predicted misconduct events are negatively associated with likelihood of being hired. Finally, we show that applicants have lower predicted complaints and disciplines, on average, when the unemployment rate is high. Taken together, these analyses provide direct evidence that the pool of applicants is stronger when outside options are scarce.

At the same time, if Florida police agencies are more constrained when unemployment rates are high, they may be more selective in their hiring decisions. To assess the role of the labor demand channel, we take advantage of additional data on the number of officers hired each year. The number of new hires is pro-cyclical in Florida, suggesting that demand for police does rise and fall with the state of the economy at large. However, we do not find evidence that agencies in Florida alter their hiring criteria across the cycle. Although passing the exam on the initial attempt is associated with an increased likelihood of being hired, we do not find that this relationship is stronger in years with higher unemployment rates. Likewise, we do not find a stronger relationship between predicted misconduct events and the likelihood of hire when economic conditions are worse. This suggests agency labor demand does fall as the unemployment rate rises, but the associated hiring decrease is not accompanied by agencies setting a higher bar for applicants.

While these results provide direct evidence that police labor supply changes across the cycle, we also conduct a bounding exercise to estimate the potential importance of the labor demand channel. With this exercise, our principal goal is to demonstrate that labor supply can account for an important share of the main effect, not necessarily that it is the sole driver of our results. Under conservative assumptions about how agencies screen out lower

quality workers, we find that labor demand responses can account for 15-30% of the overall effect of the initial unemployment rate on misconduct events.

These results suggest that when markets are weak, the supply of potential police officers increases and agencies are able to hire better officers as a result. As such, we first contribute to a growing literature on the determinants of police quality. In this context, previous papers have studied how various interventions within an agency affect the performance of officers already on the force (e.g., [Owens et al. \(2018\)](#); [Goncalves \(2021\)](#); [Owens and Ba \(2021\)](#); [Kim \(2022\)](#); [Rivera and Ba \(2022\)](#); [Mello et al. \(2023\)](#)). Within that set of papers, a few study the importance of officer pay or other financial motivations (e.g., [Mas \(2006\)](#); [Foltz and Opoku-Agyemang \(2020\)](#); [Chalfin and Goncalves \(2021\)](#)). While these papers offer valuable evidence on how to improve performance of existing police, a separate literature has developed on selection into policing and officer recruitment ([McCrary, 2007](#); [Prendergast, 2007](#); [Dharmapala et al., 2016](#); [Ornaghi, 2019](#); [Linós and Riesch, 2020](#); [Cox and Cunningham, 2021](#); [Cox et al., 2021](#)). These studies, however, do not consider the empirical relationship between outside options, officer selection, and labor supply. Our paper is one of the first to examine the role of improving the relative appeal of policing as an occupation, whether that be through financial incentives or other means, in attracting better officers. Importantly, our findings suggest that agencies attract higher quality applicants when policing is more appealing than outside options. As a result, better officers are hired and those improvements persist despite the potential standardizing effects of police culture and training. In general, recruitment, selection, and retention of public sector workers are critical policy levers, and our paper informs how government agencies can hire and retain good workers. Therefore, we build on an existing literature about the effect of financial or career incentives on the recruitment and performance of public-sector employees ([Propper and Van Reenen, 2010](#); [Ferraz and Finan, 2011](#); [Dal Bó et al., 2013](#); [Nagler et al., 2020](#); [Bertrand et al., 2020](#)).

Furthermore, our findings contribute to an extensive literature on the cyclicity of labor market flows, and how this differs by worker skill and across the public and private sectors ([Krueger, 1988](#); [Kopelman and Rosen, 2016](#)), as we show that the applicant pool of police officers changes across the cycle, and so does their on-the-job performance. By showing that the applicant pool for policing in Florida improves during recessions, we also contribute to a literature that studies how the quality of the unemployment pool varies across the business cycle ([Pries, 2008](#); [Elsby et al., 2015](#); [Mueller, 2017](#)). Our results also fit in with recent work by [Crane et al. \(2023\)](#) showing counter-cyclical upgrading of workforce quality during recessions, and with [Hirsch et al. \(2018\)](#)'s result showing that employers' monopsony power increases during downturns, thus weakening workers' outside options. We further connect to a related literature that explores how the quality of new teachers and nurses responds to

outside options in the labor market (Propper and Van Reenen, 2010; Britton and Propper, 2016; Nagler et al., 2020; Deneault, 2023).⁴

Most closely related to our work is Nagler et al. (2020)’s paper on the relationship between economic conditions and teacher quality. We build on their creative approach to understanding public sector labor supply in a number of ways. First, we document similar patterns for police, a separate, large public sector occupation. Policing has lower entry costs with respect to training and credentials than teaching, which may make it particularly responsive to economic conditions. Second, we employ data that includes each year of an officer’s employment spell from 1985 to 2016, allowing us to observe and analyze attrition directly. With this data we can also estimate dynamic impacts on officer quality, rather than the effect on quality at a single point in time. Lastly, we use rich data on entry exams and on the number of officers hired to explore the labor supply and demand mechanisms, respectively.

2 Data, Institutional Context and Methodology

2.1 Data Sources

Our empirical analysis relies on administrative data from the Florida Department of Law Enforcement (FDLE) (FDLE, 2016; Goncalves, 2021; Dharmapala et al., 2022). In particular, we use data on all employment spells for police officers in Florida from 1985 through 2015, which include information on the start and end date for each spell, the reason for separation, the employing agency, and various outcomes related to police quality. We also observe the officer’s name, birth year, gender, and race. Following Grunwald and Rappaport (2019), we restrict our analysis to full-time law-enforcement officers. Moreover, we drop a small number of officers who started their employment at two different agencies at the same time.

Importantly, FDLE data includes multiple measures of officer performance. Specifically, we use FDLE administrative data covering misconduct events and disciplinary actions held against officers. Misconduct events included in the FDLE data encompass civilian or internal affairs complaints previously sustained by a local law enforcement agency.⁵ The FDLE then opens a ‘complaint’ file and starts a disciplinary investigation — these FDLE-level investigations comprise the agency-sustained complaints considered in our analysis. Complaints

⁴Our paper also adds to a growing literature on the relationship between economic conditions at career start and subsequent labor market outcomes (Oyer, 2008; Kahn, 2010; Oreopoulos et al., 2012; Böhm and Watzinger, 2015; Altonji et al., 2016; Schwandt and Von Wachter, 2019).

⁵Local agencies sustain a misconduct event if the purported offense by the officer violates a “good moral character” requirement. If the offense is sustained, the local agency must report its findings to the FDLE.

are evaluated by an FDLE commission and if the commission finds ‘clear and convincing’ evidence of officer misconduct, it can discipline the officer following existing guidelines (Dharmapala et al., 2022). We follow Grunwald and Rappaport (2019) and Dharmapala et al. (2022), among others, and use complaints and disciplines as proxies for officer quality. Close to 30% of complaints result in a disciplinary action in the FDLE data.

We also rely on information in the FDLE data on the reason for officer separations. We consider whether an officer was fired from their position as a proxy for officer quality, motivated by the idea that police agencies are more likely to fire officers with poor performance. Moreover, since FDLE data includes the specific reason for which an officer was fired, we also analyze whether an officer was fired for directly ‘violating agency policy,’ which may more closely capture events in which an officer was fired due to poor performance on the job. In our data, among the sample of fired officers, 62% of them were dismissed for violating agency policy.⁶ These outcomes capture officers at increasingly worse parts of the quality distribution, allowing us to observe effects for a range of police misconduct. At the other end, we also observe whether officers separate from their agency voluntarily, a potential indication that they are able to find better outside options once economic conditions improve.

Our main variable for economic conditions at career start comes from annual variation in the national unemployment rate.⁷ We follow the literature on the impacts of graduating in a recession and use the unemployment rate as our preferred measure of business cycle conditions as it captures the extent to which potential police officers’ outside options may be limited. We discuss this choice and robustness to using alternative measures of economic conditions, such as the Florida-wide unemployment rate, in Section 3.2. In supplementary analyses, we also use information on county-level unemployment rates (available since 1990) aggregated to seven sub-state regions designated by the FDLE. Altogether, we merge the measures capturing economic conditions to each year of employment for all officers included in our sample.

2.2 Institutional Background

Becoming a police officer in Florida involves multiple steps. First, candidates must meet a number of minimum requirements, including being at least 19 years old, having a high school degree or GED, and passing a background check. Moreover, they must pass an exam evaluating basic competencies and a physical examination. Upon meeting these

⁶Among the 38% of officers fired for other reasons, 70% failed to complete required training programs, 16% left for performance-related issues, 12% for administrative separations, and 1% for other reasons.

⁷We use the unemployment rate faced by an officer at the start of each employment stint, except when the previous stint resulted in a transfer within agency. For those officers, we use the unemployment rate at the start of their careers. Results are robust to alternative definitions.

requirements, candidates must complete the Basic Recruitment Training Program, which prepares them for becoming an officer through academic courses and physical preparation, amounting to a total of 770 hours. After the training course (or if they have previously served as police officers either in Florida or in other states), applicants must pass the State Officer Certification Exam (SOCE), which evaluates their preparation to become police officers. In general, upon passing the exam, candidates apply to a police agency, and hiring is done at the discretion of the agency.⁸ In the FDLE data, 59% of applicants were hired within one year after passing the SOCE.

We use the FDLE data to assess the characteristics of new police officers in Florida. New hires are on average 29 years old, and close to 20% of officers are older than 35 at their career start. The vast majority of new officers in Florida are male (86%) and white (76%), see Table A1. To understand officers' occupational trajectories, we take advantage of the longitudinal component of the Current Population Survey in 1985-2015.⁹ We identify all individuals who transitioned into or out of policing and examine their occupational transitions. A substantial share of officers are previously employed in other public and private protective service occupations (22% and 10%, respectively), but an important share come from non-employment (16%) or from more distinct occupational categories like administrative support (10%), management (7%), business (6%), or sales (4%), see the second panel of Table A1. Likewise, individuals transition to a similar set of occupations after their employment as police. This evidence indicates that officers move across a variety of occupations, suggesting they may adjust their labor supply in response to economic conditions.

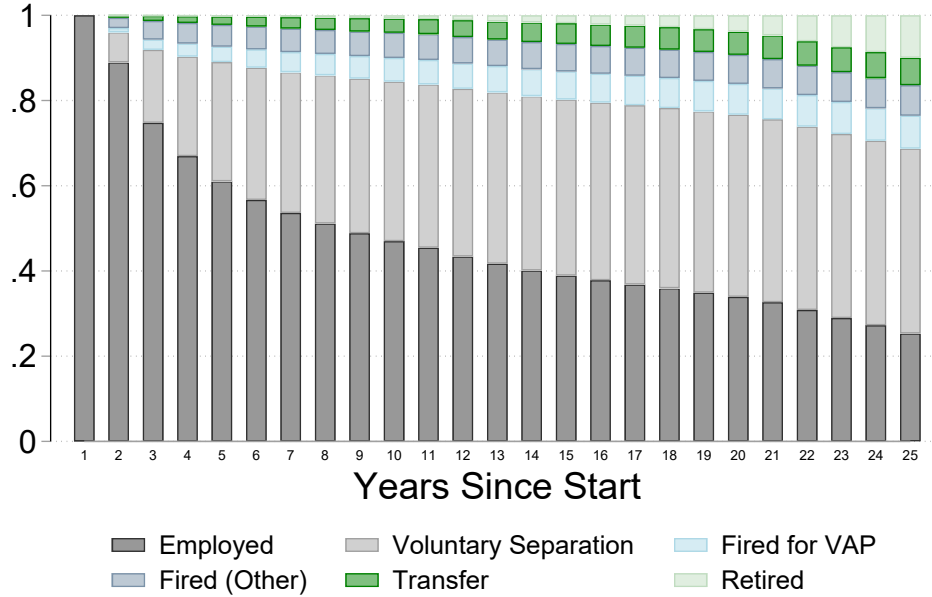
We turn back to the FDLE data to describe long-term employment outcomes for a group of young officers hired in 1985-1989 (Figure 1). First, a sizable share of officers leave their jobs early into their careers, with 61% remaining employed after five years and 25% after twenty-five years. The majority of officers who leave within ten years do so voluntarily, with smaller shares being fired for violating agency policy or fired for other reasons. Figure A1 further shows that despite a small increase in retention rates for more recent entry cohorts, around 60% of officers remain employed in FDLE five years after being hired.

External Validity. An important advantage of our data is that it covers all agencies in the state of Florida rather than a single large urban department. Studying these patterns in Florida alone is substantively important because it has the nation's fourth largest police workforce, representing 6.5% of police employment in the U.S. since 1980. A potential

⁸We describe the general process for becoming a police officer in Florida, but specific agencies may augment the process or add requirements.

⁹The CPS is limited to the civilian, non-institutional population, meaning direct transitions from active-duty military service are not observed (U.S. Census Bureau, 2020b).

Figure 1: Employment Outcomes by Years since Hire



Notes: This figure presents employment outcomes for a cohort of officers younger than 35 at hire who started between 1985 and 1989. We show the share of workers who remained employed 25 years after starting, as well as the proportion who separated voluntarily, were fired for violating agency policies, fired for other reasons, transferred to other agencies or eventually retired.

drawback, however, is that the labor market for police officers in Florida may differ in important ways from the labor market for police in other states. To explore this concern, we compare the firing and separation rates in Florida to all other states in the U.S. In Figure A2, we draw on information from a 2008 report by the Bureau of Justice Statistics, on data from the 2003 LEMAS, and the CPS for 1985-2015, where we find that Florida has firing and separation rates that are similar to nationwide rates and fall approximately in the middle of the state-level distribution.¹⁰ The similarity across these features of the police labor market suggests that estimates from Florida may be more broadly informative across the country. It bears emphasis that policing in Florida is interesting in its own right given its large share of overall police employment in the United States.

2.3 Summary Statistics

Our sample includes 103,971 unique employment spells, which last an average of 9 years. We present summary statistics in Table 1. The first column presents average statistics for our

¹⁰Within Florida, agencies that have unionized at some point before 2019 have lower firing rates than agencies that have not unionized. However, we show in Section 3.4 that our main results emerge for new hires in both unionized and non-unionized agencies.

Table 1: Summary Statistics

	Full Sample	Above Median UN at Start	Below Median UN at Start
Complaints	0.070 (0.256)	0.061 (0.240)	0.077 (0.267)
Disciplinary Actions	0.020 (0.141)	0.018 (0.133)	0.022 (0.147)
Fired	0.094 (0.292)	0.086 (0.281)	0.100 (0.301)
Fired for Violating Agency Policy	0.063 (0.243)	0.057 (0.232)	0.067 (0.250)
Voluntary Separation	0.540 (0.498)	0.539 (0.499)	0.542 (0.498)
Unemployment Rate in Starting Year	5.936 (1.332)	7.116 (1.145)	5.041 (0.521)
Observations	103,971	44,833	59,138

Notes: This table produces summary statistics for the sample of full-time police officers beginning an employment spell from 1985-2015. We present the mean for the main variables of interest along with the standard deviations in parentheses. Column 1 shows statistics for all officers, column 2 limits to officers who start in a year with an above median unemployment rate (UN), and column 3 limits to officers who start in a year with a below median unemployment rate. Officers may appear more than once in the data if they have multiple employment spells. The rows include the fraction to receive any complaint during their spell, fraction to receive any disciplinary action in the spell, fraction fired in their spell, fraction fired for violating agency policy in their spell, fraction who separate voluntarily, and the mean unemployment rate in the starting year.

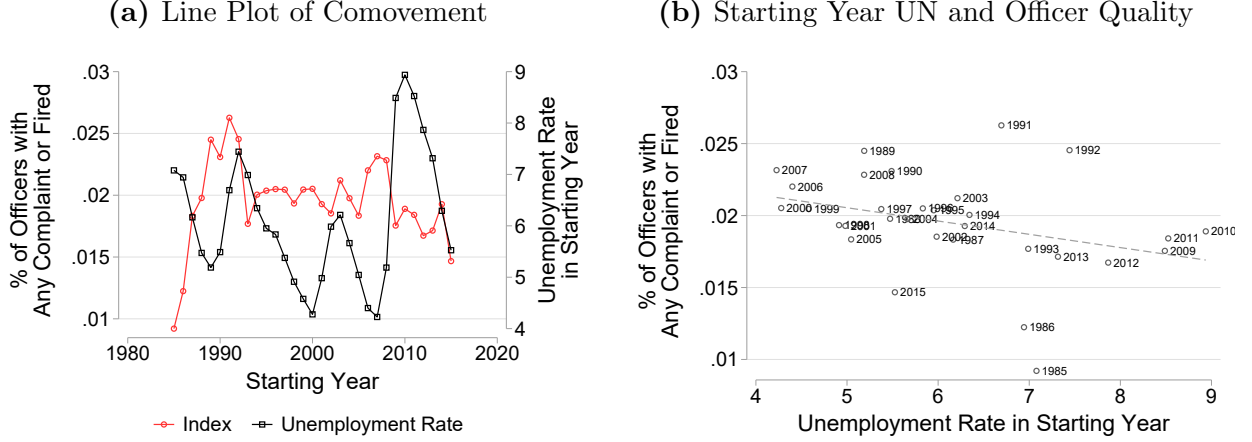
sample. 7% of employment spells involve a complaint against the officer, and close to 30% of these complaints eventually result in disciplinary action. On average, 9.4% of employment spells end in the officer being fired from their job, and two-thirds of those fired are fired for violating agency policy. A sizable share of employment spells (54%) end in voluntary separation. In the remaining columns, we split the sample by whether the employment spell began in a year above or below the median unemployment rate (5.8%), where the difference in unemployment rates at start across these two groups amounts to 2.1 percentage points. Employment spells that started in high-unemployment years are less likely to involve a complaint (6.1% vs. 7.7%), less likely to be fired (8.6% vs. 10%), and less likely to be fired for violating agency policy (5.7% vs. 6.7%). The raw data also shows that employment spells starting in high unemployment years are less likely to receive complaints or be fired than those starting in low unemployment periods (Figure 2).¹¹

2.4 Empirical Strategy

We examine the impact of economic conditions at career start on police officer quality by estimating the effect of the unemployment rate at the time of hire on our main officer

¹¹In Figure A3, we present the co-movement plots for each outcome separately.

Figure 2: Starting Year Unemployment Rate and Officer Quality, Residualized



Notes: Panel (a) and panel (b) plot the relationship between a summary measure of officer quality and unemployment rate, after residualizing both using current year unemployment and agency fixed effects. Panel (a) shows how these two variable move with each other over time. Panel (b) plots the two against each other and overlays a linear fit. At the starting year level, unemployment rate in starting year explains 11.6% of the residual variation in likelihood an officer receives a complaint or is fired in their first five years.

outcomes in the following equation:

$$Y_{iat} = \alpha + \gamma UR_i^S + \beta UR_t + \theta_a + \varepsilon_{iat} \quad (1)$$

where Y_{iat} represents an outcome of interest for officer i employed in agency a in year t ; UR_i^S captures the national unemployment rate that prevailed in the year in which the officer was first hired. θ_a capture agency fixed effects which account for differences in outcomes across police agencies in Florida. In equation (1), we control for contemporaneous unemployment (UR_t) implying that γ allows us to recover the effect of the initial economic conditions net of all subsequent unemployment rates (Von Wachter, 2020). This accounts, for example, for the possibility that officers may adjust their policing behavior due to facing weak concurrent outside options. We additionally estimate a specification that accounts for potential changes in the policing environment by including agency-by-year fixed effects (θ_{at}) in equation (1).

To implement equation (1), we need to account for the fact that a sizable share of officers leave their jobs early into their careers. As such, we could estimate the equation including officers' entire employment stints — implying that the latter years would only include a few individuals — or focusing early in their careers, yet missing potential longer-term impacts. In our main specification, we take a middle ground approach and focus on officers' outcomes in their first five years since hire, such that our main estimation sample is an officer-year panel that tracks each employment spell for up to five years. Within this panel, we record

complaints and disciplines in the year they occur. Separation events, such as being fired or separating voluntarily, are coded in an officer’s final year of observation for that spell, which concludes their contribution to the sample. Our main estimation sample thus includes all police officers who were hired between 1985 and 2015, and we observe their outcomes through 2016.¹²

We conduct an extensive set of robustness tests along other dimensions as well. First, we re-estimate equation (1) without including the contemporaneous unemployment rate, which allows us to recover the total impact of initial conditions on police quality. In addition, we assess the robustness of the results to the inclusion of time trends and to multiple leads and lags of the unemployment rate. We estimate the results using the various alternative measures of the unemployment rate, restricting our sample to stints starting before 2011 (to observe outcomes fully through the end of our data) and analyzing outcomes through the first two years of employment. We also estimate a specification in which we control for officer characteristics at career start and specifications in which we control for officers’ experience or within-agency tenure. We also estimate a version of equation (1) that controls for current and starting year crime rates. These results and others are described in more detail in Section 3 and in their respective table and figure notes.

We further analyze the dynamic impacts of initial economic conditions on officer outcomes by estimating equation (1) separately for each year s since the officer was hired, where s ranges between zero and five. This specification allows us to present novel evidence by tracing out the dynamic effects of unemployment rates at career start on officer quality across their first five years of employment.

3 Results

3.1 Main Results

In Table 2, we present the estimates from equation (1). In the first column, we show that officers hired when the unemployment rate is one percentage point higher receive 0.00077 fewer complaints per year, or 8.5% of baseline complaints, in the first five years of employment.¹³ As such, an officer who is hired during a period with an above-median unemployment rate is 20 percent less likely to receive an agency-sustained complaint against them compared to a counterpart hired during below-median periods. This is similar in magnitude to civilian

¹²Our results are robust to restricting the sample to officers hired between 1985 and 2011, for whom we observe the full five-year outcome window, as discussed in Section 3.3.

¹³In Table A2, we show summary statistics on the baseline prevalence of complaints and disciplinary actions across both the first five years and the full employment stint for officers in our sample.

complaint reductions caused by increased police oversight (Jordan and Kim, 2022; Rivera and Ba, 2022), procedural justice training (Wood et al., 2020), and the introduction of body-worn cameras (Kim, 2022). We find similar impacts on the prevalence of disciplinary actions, as a one percentage point increase in the unemployment rate at career start results in 0.00039 fewer disciplinary actions per year, or 13% of baseline disciplinary actions, in the first five years.¹⁴ In our view, an effect of this magnitude is meaningful, as we study agency-sustained misconduct events, which are likely correlated with more common, but harder to observe, behaviors related to police quality.

Table 2: Effect of Starting Year Unemployment Rate on Officer Quality

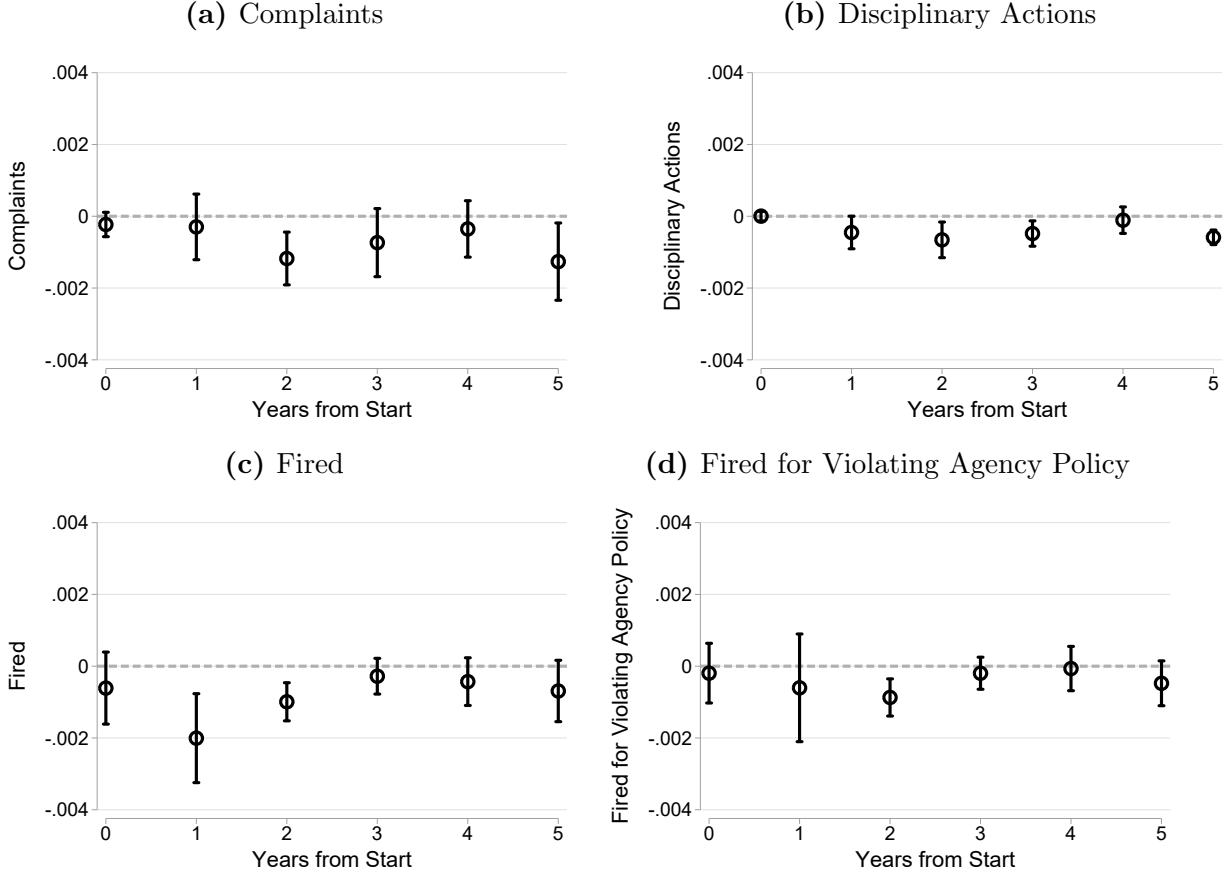
	Complaints	Disciplinary Actions	Fired	Fired for VAP
	(1)	(2)	(3)	(4)
Unemployment Rate in Starting Year	-0.00077*** (0.00027)	-0.00039*** (0.00013)	-0.00073** (0.00027)	-0.00044** (0.00017)
Constant	0.0109*** (0.0015)	0.0031*** (0.0006)	0.0181*** (0.0025)	0.0097*** (0.0016)
Observations	468,924	468,924	468,924	468,924
R^2	0.0028	0.0007	0.0043	0.0034

Notes: This table displays results estimated from equation (1). Standard errors clustered at the level of officer’s starting year in parentheses. Each observation corresponds to an officer-by-year employment record. Column 1 estimates the relationship between the unemployment rate in the officer’s starting year and the number of complaints on their file from years 0-5 since the starting year. Column 2 estimates the relationship between starting year unemployment rate and disciplinary actions. Column 3 estimates the relationship between starting year unemployment rate and likelihood of being fired. Column 4 estimates the relationship between starting year unemployment rate and likelihood of being fired for violating agency policy. All columns limit the sample to years 0-5 from each officer’s starting year. All specifications include a control for current year unemployment rate and agency fixed effects. Table A2 reports the overall mean of each of these outcomes within the first five years of an officer’s career. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

In the last two columns of Table 2, we present the estimated impacts on the likelihood of being fired. We find significant impacts of the initial unemployment rate on the likelihood that officers are subsequently fired and that this firing is due to violating agency policy. In fact, the estimated coefficients for both variables are such that a one percentage point increase in the initial unemployment rate results in a 4.9% decrease in the likelihood of being fired altogether and of being fired for violating agency policy in the first five years of employment.

¹⁴To illustrate the economic significance of our results, we assess how a higher initial unemployment rate affects the total number of misconduct events for an entering cohort. The average entry cohort comprises 3,350 officers who remain employed for an average of four years. We multiply these numbers by the estimated effect on complaints and disciplines in Table 2 and find that a two percentage point increase in the unemployment rate would lead to a reduction of 20.6 complaints and 10.5 disciplines for the entire cohort. Since the average entry cohort logs 120 complaints and 36 disciplines over their first five years in the force, our estimates represent a notable 17% and 28% decrease in overall complaints and disciplines, respectively.

Figure 3: Starting Year Unemployment Rate and Officer Quality, By Year



Notes: This figure displays coefficients estimated from equation (1) estimated separately for each year since start. With each coefficient, we plot the 90 percent confidence interval, based on standard errors clustered at the level of officer's starting year. Panel (a) displays estimates of the effect of starting year unemployment rate on complaints in the year the officer starts, the year after, 2 years after, 3 years after, 4 years after, and 5 years after. Panel (b) displays the same time path of estimates for disciplinary actions. Panel (c) shows estimates for the likelihood that the officer is fired and panel (d) shows estimates for the likelihood that they are fired for violating agency policy.

In Figure 3, we present the dynamic effects of initial unemployment on officer outcomes across the first five years of employment. In the first panel, we present the estimated impacts on the prevalence of complaints. We find no significant impacts of initial unemployment rates on complaints in the year of hire, yet the effects become negative and statistically significant by the second year since hire. The coefficients remain negative through the next three years, with varying statistical significance. For disciplines, we similarly fail to find an effect of economic conditions at start in the initial year, yet the effects become negative and significant for the first three subsequent years. As such, we find that the estimated impacts of initial unemployment rates on officer quality emerge as early as the second year since starting the job, indicating the effects are not driven by attrition. In their analysis of business cycles and

teacher quality, Nagler et al. (2020) use a time-invariant measure of teacher quality (value-added). Our estimated dynamic effects thus constitute novel evidence in the literature by allowing us to understand how early police quality is revealed in light of starting conditions. In the last two panels, we show that officers hired in high-unemployment periods are less likely to be fired within one to two years of their career start.¹⁵

Measuring Officer Quality. In this context, we note that the policing literature has previously highlighted the difficulty of assessing officer quality (Rim et al., 2020; Hoekstra and Sloan, 2022). We do not aim to determine a holistic measure of officer quality, instead we posit that the outcomes we study are negative aspects of officer quality. For instance, a common concern with using complaints as a measure of quality is that officers who are more proactive on the job may receive more complaints. If that were the case, then complaints could reflect good policing performance. However, Rozema and Schanzenbach (2019) have shown that officers who receive complaints against them have a far higher likelihood of facing future civil rights litigation. Moreover, the complaints we study are agency-sustained complaints, not civilian complaints and complaints with disciplinary actions are further sustained by FDLE.¹⁶

In Section 4, we further show that a distinct measure of officer quality from the civil service exam is correlated with our outcome measures – officers who pass the entry exam on the first try are less likely to have received complaints and disciplinary actions. Lastly, 55% (86%) of officers with a complaint (discipline) in their first two years on the force are eventually fired, compared to 11% (12%) of officers who do not receive complaints (disciplines) in their early career. Moreover, 86% of officers who receive a complaint in their first two years are eventually decertified from policing in Florida, compared to just 11% of those who did not receive a complaint.¹⁷ Altogether, these results support the claim that complaints and disciplines capture negative aspects of officer quality in our setting.

Addressing Differences in the Nature of Policing. There are two distinct potential concerns related to time-varying differences in the nature of policing in our setting. The first

¹⁵Our results show that officers hired in a recession are less likely to have been fired early in their careers. Officers who are fired early in their careers are at the bottom of the quality distribution, suggesting that had they remained employed, they would have likely incurred additional complaints and disciplines.

¹⁶Our results are robust to directly controlling for the annual discipline rate (given by the share of complaints that result in disciplines), which could also vary across the business cycle (Table A3).

¹⁷We leverage additional information capturing whether officers were decertified in Florida, where we observe the decertification date for 55% of decertified officers. We estimate a specification that relates the unemployment rate at hire to whether an officer is ever decertified (Table A4). Officers hired in high-unemployment years are less likely to have been decertified (p-value = 0.166).

is that the contemporaneous policing environment may be correlated with the starting year unemployment rate in a way that confounds our main estimates. The second is that the starting year policing environment could have path-dependent effects on officer quality, and may thus be one mechanism through which our effects arise. In this section, we examine the robustness of our results to account for the fact that the contemporaneous nature of policing may vary across agencies and over time. In Section 4.3, we explore the possibility that the policing environment in one’s starting year may partially explain our results.

We first estimate a modified version of equation (1) that includes agency-by-year fixed effects, implying we compare outcomes for officers working in the same agency in the same year but who were hired at different points in the cycle. We present the estimated results in the first panel of Table 3, finding that officers in high unemployment years are less likely to have received disciplines against them, to have been fired, and to have been fired for violating agency policy. Since this specification focuses on the first five years since career start, the year fixed effects are effectively identified *only* from officers hired in the past five years.

To improve the identification of the year fixed effects, we re-estimate this specification using all employment years, and we find significant effects of initial unemployment on the prevalence of complaints, disciplines and the likelihood of being fired for violating agency policy. However, these estimates suffer from attrition bias, as a sizable share of officers leave policing early into their career.¹⁸ To identify year fixed effects from all employed officers and limit attrition bias, we estimate an additional specification in which we first residualize each outcome of interest from agency-year fixed effects, controlling for differences in the prevalence of each outcome for *all* officers employed in a particular agency in that year. We then use the residualized variable as our outcome of interest in equation (1). We present the results in the third panel of Table 3, where we find that officers hired in high unemployment years are less likely to receive complaints and disciplines, and are less likely to be fired and fired for violating agency policy. Altogether, these results indicate that our main results are robust to controlling for differences in the contemporaneous policing environment in various ways.

We also assess the robustness of our results by estimating a modified version of equation (1) that uses variation in initial local unemployment rates across seven sub-state regions and controls for agency-by-year fixed effects (Table A5). We find that officers who started with higher local unemployment rates have fewer complaints, disciplines, and a lower likelihood of being fired/fired for violating agency policy, with significant effects on the last

¹⁸The potential bias comes from the fact that including more years allows differential attrition to grow over time. If lower-quality officers exit policing at higher rates, then cohorts that differ in quality at career start will become more alike over time.

Table 3: Effect of Initial Unemployment on Officer Quality, Alternative Specifications

Panel A: Agency-by-Year Fixed Effects, Five Year Window

	Complaints (1)	Disciplines (2)	Fired (3)	Fired for VAP (4)
Unemployment Rate in Starting Year	-0.00035 (0.00028)	-0.00020*** (0.00007)	-0.00095*** (0.00026)	-0.00070*** (0.00024)
Constant	0.00828*** (0.00163)	0.00294*** (0.00036)	0.02440*** (0.00159)	0.01270*** (0.00150)
Agency-Year FE	Yes	Yes	Yes	Yes
Years Since Hire	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years
R^2	0.0155	0.0041	0.0145	0.0140
Observations	468361	468361	468361	468361

Panel B: Agency-by-Year Fixed Effects, All Years

	Complaints (1)	Disciplines (2)	Fired (3)	Fired for VAP (4)
Unemployment Rate in Starting Year	-0.00035** (0.00014)	-0.00017*** (0.00004)	-0.00028* (0.00015)	-0.00032** (0.00012)
Constant	0.01122*** (0.00096)	0.00342*** (0.00024)	0.01606*** (0.00091)	0.01036*** (0.00080)
Agency-Year FE	Yes	Yes	Yes	Yes
Years Since Hire	All Years	All Years	All Years	All Years
R^2	0.0119	0.0057	0.0166	0.0135
Observations	941108	941108	941108	941108

Panel C: Residualized Outcomes, Five Year Window

	Complaints (1)	Disciplines (2)	Fired (3)	Fired for VAP (4)
Unemployment Rate in Starting Year	-0.00026 (0.00018)	-0.00020*** (0.00006)	-0.00073*** (0.00023)	-0.00044*** (0.00014)
Constant	-0.00111 (0.00098)	0.00041 (0.00033)	0.01051*** (0.00163)	0.00298** (0.00125)
Agency FE	Yes	Yes	Yes	Yes
Years Since Hire	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years
R^2	0.0012	0.0007	0.0016	0.0007
Observations	468703	468703	468703	468703

Notes: In the first two panels of this table, we present the results from a modified version of equation (1) that includes agency-by-year fixed effects in: $Y_{iat} = \alpha + \gamma UR_i^S + \delta T_{iat} + \theta_{at} + \varepsilon_{iat}$ where θ_{at} captures agency-by-year fixed effects, and T_{iat} represents officer i 's tenure in agency a by year t . Panel A estimates this equation restricting the sample to the first five years of an officer's employment spell, whereas Panel B includes all years of employment. Panel C presents the results from a modified version of equation (1), which uses residualized outcomes as the outcome variable of interest. We residualize the four outcome variables from agency-by-year characteristics in a preliminary step in: $Y_{iat} = \alpha + \phi_{at} + v_{iat}$, and we then use the residualized outcome from this equation ($\widehat{Y}_{iat}$) as the outcome variable in equation (1). See Table 2 notes for a description of the outcome variables. Standard errors are clustered at the officer's starting year level in parentheses in all specifications. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

three outcomes.¹⁹ In this context, we also leverage spatial variation in the severity of the Great Recession across Florida in a difference-in-differences design, following Yagan (2019) and Weinstein (2025). Officers hired in harder-hit commuting zones after the recession have fewer complaints, disciplines, and involuntary separations (Table A6), although the confidence intervals on these estimates are often very wide. The event study presented in Figure

A4 shows that quality was not trending downward in the severely affected areas in the years preceding the Great Recession, but begins to decline with its onset. We do not use local unemployment rates in our main specifications because it requires us to focus on officers hired since 1990, it does not account for potential within-Florida migration flows, and it requires using local unemployment estimates that are inherently noisier than national-level estimates. Nevertheless, these results are consistent with the evidence presented in Table 2.

Agency-Level Public Safety Outcomes. Our main results focus on four outcomes that are negative aspects of officer quality. To our knowledge, there is no empirical evidence supporting the concern that officers who are worse along these dimensions are, on the other hand, better at deterring crime or clearing cases. Nonetheless, we explore that possibility to the best of our ability using variation in agencies’ employment composition, as we do not observe officer-level arrest data. In particular, we analyze whether agencies with a larger share of officers hired in high-unemployment years attain better public safety outcomes. We capture agencies’ annual outcomes using Uniform Crime Report (UCR) data on agency-level crime rates and clearance rates between 1985-2015. We study how these outcomes vary depending on the composition of officers in an agency, including average race, age, gender and unemployment rate at career start of all officers employed in the agency in that year. We control for agency size, contemporaneous unemployment, as well as agency and year fixed effects. Agencies with a larger share of officers hired in high unemployment years have lower overall, violent, and property crime rates and higher clearance rates (Table A7).

3.2 Alternative Unemployment Rate Measures

Our preferred specification follows the related literature (e.g. Nagler et al. (2020)) by relying on cross-year variation in national economic conditions. We take this approach for two primary reasons. First, unemployment rates estimated at sub-national levels are inherently noisier than the national rate. Second, individuals in the labor market for police in Florida originate from a wide range of states. We expand on these points below; however, we also show the robustness of our results to using a variety of alternative measures of the unemployment rate.

As noted above, we use the national unemployment rate instead of Florida-wide measures to reduce measurement error in our main variable of interest. For instance, the 90% confidence interval around the national unemployment rate in 2022 spanned just 0.22 percentage points, whereas the corresponding confidence interval for Florida was nearly three times as

¹⁹In the second panel of Table A5, we re-estimate this specification using all employment years and find significant effects across all four outcomes.

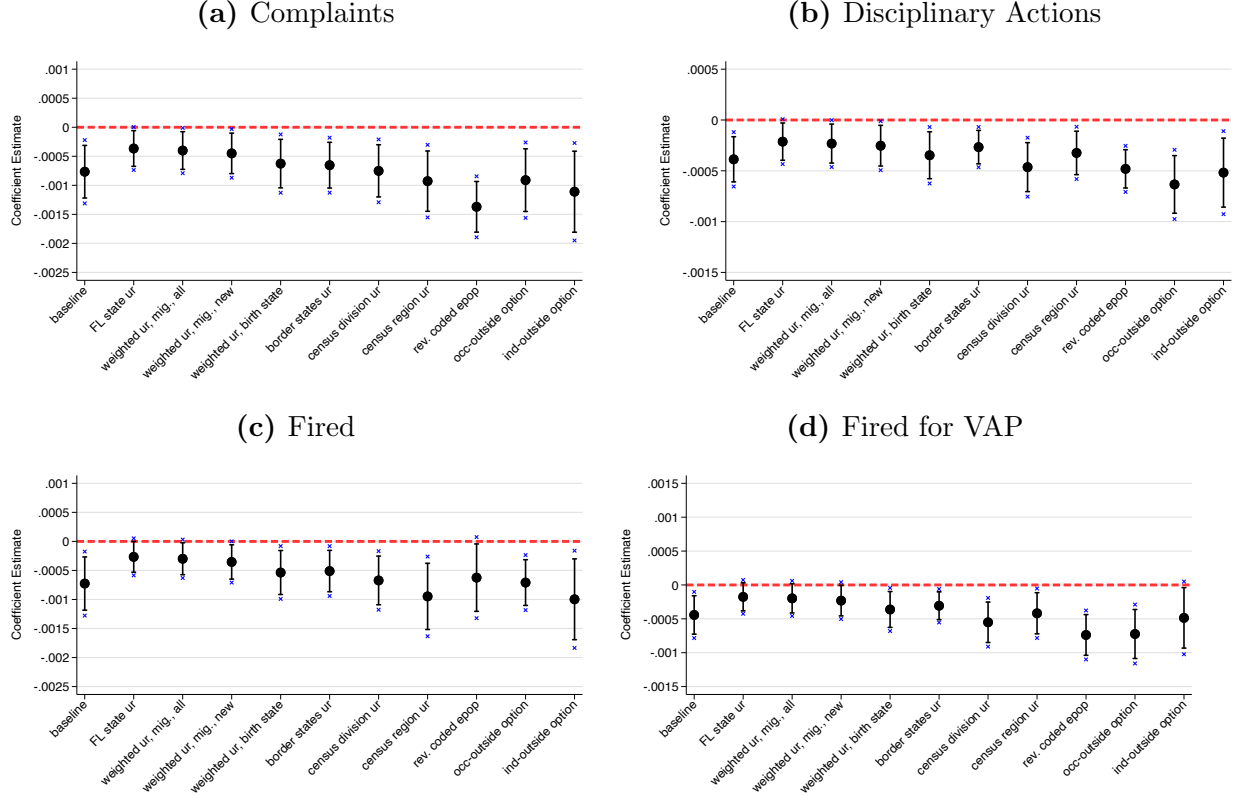
large (U.S. Bureau of Labor Statistics, 2023a,b). The greater imprecision in state-level unemployment rate estimates arises because these measures are based on smaller samples and rely on modeling that incorporates auxiliary data sources, which introduce additional measurement error (Tiller, 2006). As a result, using state-level unemployment measures in lieu of the national unemployment rate may lead to attenuation bias in the estimated impacts of initial economic conditions on officer outcomes.

Furthermore, the labor market for police in Florida is national in scope. We characterize this fact in three different ways. First, using decennial census data from 1980-2000, we estimate that 12% of all police officers in Florida lived outside the state five years prior. This statistic is very likely to understate the share of *new* police in Florida who lived outside of Florida five years ago. In fact, when we examine police under the age of 30 in Florida, 17% lived outside of Florida five years ago. We further draw on the 1970 decennial census, which asks about current and prior occupation. Among new protective service employees in 1970, 25% lived outside of Florida five years ago. Finally, even these statistics may understate the degree to which potential police in Florida consider outside options in other states. The 1980-2000 decennial censuses indicate that approximately 70% of Florida’s police officers are born out of state. These results show that the Florida-specific unemployment rate does not comprehensively capture economic conditions and outside options in our setting.

Despite our rationale for focusing on the national unemployment rate as the primary measure of variation in outside options, we show robustness to a variety of alternative measures. The results of these robustness checks are in Figure 4. The first estimate shows our baseline result using the national unemployment rate. The second estimate uses the unemployment rate in Florida alone. Since this measure does not comprehensively reflect outside options for officers and introduces additional measurement error, we expect it to attenuate our estimates. Our results confirm this prediction, yet the point estimate is of the same sign and similar statistical significance ($p\text{-value} = 0.053$ for complaints) as our main estimates. We then estimate three specifications motivated by the officer migration statistics discussed above.²⁰ The next three columns use the average unemployment rate from Florida and geographically close states: border states, states in the same Census division, and states in the same Census region. We also use the nationwide employment-to-population ratio, reverse-coded to have the same sign as our unemployment results. Our main results are robust to all of these alternatives and do not depend on our choice to focus on the national unemployment rate.

²⁰In particular, we first calculate a weighted unemployment rate based on the share of Florida officers who lived in different states five years prior and those states’ respective unemployment rates. We then use the corresponding share of new protective service workers in Florida by state of residence five years ago. Third, we use the share of all officers in Florida by state of birth.

Figure 4: Effect of Starting Year Unemployment Rate on Officer Outcomes, Alternative Unemployment Rate Measures



Notes: This figure shows robustness tests for our main results. In the following order, it shows: the baseline estimates (Table 2), the Florida-specific unemployment rate, the next three estimates use unemployment rates weighted by the share of all officers in Florida living in that state five years ago, by the share of new protective service workers in Florida by state of residence five years ago, and by the share of all officers in Florida by state of birth, respectively. The next three estimates use the average unemployment rate from Florida and nearby states: bordering states, same Census division, and same Census region. Finally, we use the nationwide employment-to-population ratio (reverse coded), and an occupation (industry)-weighted unemployment rate based on new officers' origin occupations and industries (from CPS data). We plot the 90% confidence interval, based on standard errors clustered at the level of the officer's starting year, along with 95% intervals per estimate in blue.

To more closely capture the relevant outside options for new police officers, we additionally construct weighted unemployment rates based on officer transitions from other occupations/industries. We use the occupation transition rates in Table A1, and calculate the unemployment rates in these occupations following Bureau of Labor Statistics (2015b), and in turn follow the same approach at the industry-level (Bureau of Labor Statistics, 2015a). By combining the occupation-level annual unemployment rate with the share of new police officers coming from various occupations/industries, we create an alternative proxy for the unemployment rate that may better reflect the outside options for new officers. The last two estimates in Figure 4 show the results of this analysis, indicating that our main findings are robust to these alternative measures of outside options.

3.3 Additional Robustness Checks

We test the robustness of our main results to alternative specifications in Figure A5. First, our main results are robust to removing contemporaneous unemployment as a control, implying that initial unemployment affects officer quality regardless of subsequent economic conditions. All results remain significant when including linear time trends in the starting year, which accounts for potential secular changes in hiring practices over time. Furthermore, our results are robust to including agency-specific linear trends, allowing for heterogeneity in how hiring practices evolved across different agencies over time. The effects on complaints and both firing outcomes remain similar even when conditioning on various leads and lags of the initial unemployment rate. Our results are also robust to focusing on officers who start their jobs in 1985-2011, to controlling for worker observables at baseline, and to controlling for current and starting year crime rates.²¹ We also include specifications in which we control for officer’s experience or within-agency tenure—despite introducing these endogenous controls, we find similar results. Furthermore, we find larger estimated coefficients when focusing on officers’ first two years since hire, which is consistent with the dynamic impacts presented in Figure 3.^{22,23} Lastly, our results are also robust to restricting the estimation sample to each officer’s first employment spell (Table A13).

To assess the robustness of our results to alternative time windows, we estimate the survival function of police officers receiving a complaint or discipline by the number of years since hire. We compare the survival function of the time until an officer receives a complaint/discipline by whether the officer was hired in an above- or below-median unemployment year (Figure A7). Officers hired in low-unemployment years are more likely to receive complaints early in their careers than those starting in high-unemployment periods, with similar results emerging for disciplines. This, in addition to our results that include all years of an officer’s employment spell, further highlights that our findings are robust to alternative time windows.²⁴

²¹Given the rise in crime in the mid-1980s to early-1990s and the subsequent fall throughout the 1990s (Levitt, 2004), we also show robustness to excluding this early period. Table A8 shows that our main results are unaffected by excluding 1985-1994.

²²Our results are robust to using monthly unemployment rates, and to using lagged measures of initial monthly unemployment (Table A9). Our results are robust to dropping any given entry cohort or consecutive pair of cohorts from the sample, as shown in Figure A6 and Table A10.

²³In all of the main specifications, we cluster standard errors at the level of the officer’s starting year. We also consider alternative methods of statistical inference with small clusters. First, results remain unchanged when using a wild cluster bootstrap procedure (Table A12). Second, we construct placebo estimates by randomly assigning starting years to entire starting cohorts and to individual officers (Figures A8 and A9). In both cases, our estimates are outliers relative to the distribution of placebo estimates. In Table A11, we also show that our conclusions are robust to clustering at the agency-level or at the officer-level.

²⁴We find the same results when controlling for agency fixed effects and officers’ observable characteristics.

3.4 Labor Market Dynamics

In Table 4, we present estimates of equation (1) separately by officers’ age to analyze whether heterogeneous impacts emerge for younger versus older officers. We find significantly larger effects for young officers (hired when younger than 35) than for their older counterparts. The estimated impacts are statistically different across the two age groups across complaints, disciplines, and the two fired outcomes, suggesting that the quality of younger officers exhibits far stronger responses to business cycle conditions.^{25,26} This is consistent with Forsythe (2022), who shows that during recessions, the hiring rate for young workers falls faster than for older workers. As such, young workers face weaker outside options during slack labor markets, leading to stronger sorting responses into policing than for their older counterparts.^{27,28} We also explore whether the demographic characteristics of hires vary based on the unemployment rate, but do not find any evidence to this end (Table A18).

Table 4: Effect of Starting Year Unemployment Rate on Officer Quality by Age

	Complaints (1)	Disciplines (2)	Fired (3)	Fired for VAP (4)
Unemployment Rate in Starting Year	-0.00041 (0.00028)	-0.00024*** (0.00009)	-0.00002 (0.00034)	-0.00006 (0.00017)
Young	0.00939*** (0.00213)	0.00450*** (0.00118)	0.02040*** (0.00350)	0.01280*** (0.00246)
Young $\times$ UR Start	-0.00080** (0.00036)	-0.00035* (0.00019)	-0.00163*** (0.00052)	-0.00094** (0.00035)
Constant	0.00651*** (0.00159)	0.00108* (0.00055)	0.00868** (0.00340)	0.00392*** (0.00135)
Current UN	Yes	Yes	Yes	Yes
Agency FE	Yes	Yes	Yes	Yes
Years Since Hire	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years
R^2	0.0034	0.0012	0.0063	0.0048
Observations	468921	468921	468921	468921

Notes: This table displays results estimated from equation (1) including a binary variable equal to one if the officer is below 35 upon starting (i.e., young) and an interaction between that variable and the starting year unemployment rate. Standard errors clustered at the level of the officer’s starting year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

To provide further evidence on initial unemployment rates and subsequent labor mar-

²⁵This result is robust to defining age at start linearly and for officers under 30 at start (Table A14).

²⁶Table A15 presents effects by race and gender, respectively. We do not find statistically significant differences across outcomes by these characteristics.

²⁷Table A16 shows that the estimated effects of initial unemployment on officers’ complaints, disciplines and the likelihood of being fired are similar for officers employed in unionized and non-unionized agencies.

²⁸We additionally examine heterogeneity by agency size (Table A17). We re-estimate our main specification with inverse agency-size weights, and find larger coefficients across all four outcomes. We also estimate an interacted specification for large and small agencies, and similarly find larger effects for smaller agencies.

ket dynamics, we estimate equation (1) and find that, conditional on the contemporaneous unemployment rate, officers hired in high-unemployment years are somewhat more likely to voluntarily separate soon after joining the force, with significant effects in their second year since hire (Figure A10). Interestingly, when we do not control for contemporaneous unemployment in this analysis, the estimated coefficient becomes insignificant and is close to zero.²⁹ The difference in the estimates is consistent with officers remaining on the force in response to their outside options. Since unemployment rates are correlated over time, officers hired in recessions are unlikely to voluntarily separate when outside options remain limited.³⁰ However, once we account for their outside options — by controlling for contemporaneous unemployment — we find that officers hired in a recession are more likely to subsequently leave voluntarily. Altogether, since higher-quality officers are more likely to have strong outside options in subsequent years, the estimated effect on voluntary separations is consistent with our earlier results.³¹

4 Mechanisms

We have so far shown consistent evidence that the quality of police officers is significantly higher for officers hired in high-unemployment periods. This result could emerge through different channels. First, recessions may lead to large employment losses in related industries in which potential officers may find work. Thus, during economic downturns, high-quality workers may choose to seek employment as police officers in lieu of lower-paying or hard-to-find alternative jobs. At the same time, a recession may lead to a reduction in police department budgets, which would reduce the number of officers agencies could hire. In this case, our findings could arise through police departments selecting the highest-quality officers among the applicant pool. A third possibility is that the nature of policing could vary in a downturn vis-à-vis an expansion, and that starting under different conditions may have a path-dependent effect on quality. In fact, these mechanisms could be taking place at the same time in our setting. Importantly, we do not need to rule out the last two channels to show that labor supply changes contribute to our findings. We provide direct evidence on the size and quality of the applicant pool. We then assess the potential importance of the

²⁹Voluntary separation rates are similar for officers hired in high or low-unemployment years (Figure A11).

³⁰We find no evidence that officers who passed the SOCE on the first try (a proxy for officer quality) are differentially likely to separate or be fired based on the unemployment rate at career start (Table A19).

³¹From a policy perspective, we do not take a stance on whether it is preferable for agencies to attract higher quality officers with slightly higher turnover or lower quality officers with slightly lower turnover. Since the unconditional correlation is zero and the relationship conditional on the current unemployment rate is small (i.e., approximately 2% of the baseline mean), we do not expect differential voluntary separation to be a primary concern in our setting.

other channels to paint a comprehensive picture of the mechanisms driving our results.

We remark that our paper aims to answer two main questions. First, whether quality gains from recruitment persist into policing, given potential concerns that police training, culture, and the job itself could negate those gains. Addressing this does not require separating labor demand from labor supply factors. Second, we assess whether financially-motivated applicants exhibit higher quality in various dimensions, given the longstanding debate about extrinsic versus intrinsic motivation for government work.³² Our goal is to show that labor supply plays a non-trivial role behind our estimated results, not that it is the only mechanism at work.

Sample Restrictions. To provide direct evidence on these mechanisms, we rely on information from a dataset comprised of all entry-level State Officer Certification Exams taken by police applicants in Florida from 1996-2015. As discussed in Section 2, these exams are taken after applicants have finished basic recruit training, and passing the SOCE exam is a requirement to be eligible to be hired by a police agency. This data allows us to directly examine important characteristics of the quality of the applicant pool before agencies make their final hiring decisions.

To analyze the importance of the posited mechanisms, we impose various sample restrictions to the specific question at hand. First, to assess general trends in the size and quality of the applicant pool, we use the universe of all 64,998 SOCE test-takers between 1996 and 2015 (Section 4.1.1). Second, to examine whether officers who pass on the first try are more likely to be hired, we in turn restrict the sample to the 61,285 applicants who eventually passed and were thus eligible to be hired. In the same exercise, we analyze whether passing on the first try is associated with better outcomes for applicants who were actually hired in a sample of 42,032 officers.

To provide further evidence of the labor supply channel, we then present a complementary machine learning exercise that predicts misconduct events for applicants for whom we observe police academy training data (Section 4.1.2). Since this analysis aims to measure the quality of the overall applicant pool, we include all 55,189 test-takers with this training data, rather than restricting the sample to eventual passers. Lastly, to test the labor demand channel, we focus on the 52,577 eventual passers with valid training information, as this group represents the final pool of eligible candidates from which agencies actually hire and for which we can predict outcomes (Section 4.2). This exercise allows us to directly test whether agencies alter their hiring criteria across the cycle.

³²We argue that individuals who enter policing due to better wages or due to better employment prospects are both financially (or extrinsically) motivated.

4.1 Labor Supply

As noted above, labor supply responses may contribute to our results if the relative attractiveness of policing increases during economic downturns. To this end, we use Florida Criminal Justice Agency Profile (CJAP) data in 2011-2019 to examine how posted entry salaries in police agencies and sheriff’s offices vary across the business cycle. In Table A20, we show that starting salaries across police agencies do not vary across the business cycle, which stands in contrast with extensive evidence showing that wages of private sector hires are strongly pro-cyclical (Bils, 1985; Martins et al., 2012; Carneiro et al., 2012; Gertler et al., 2020). As such, these results suggest that becoming a police officer becomes relatively more attractive during high unemployment periods.³³

4.1.1 Entry Exams

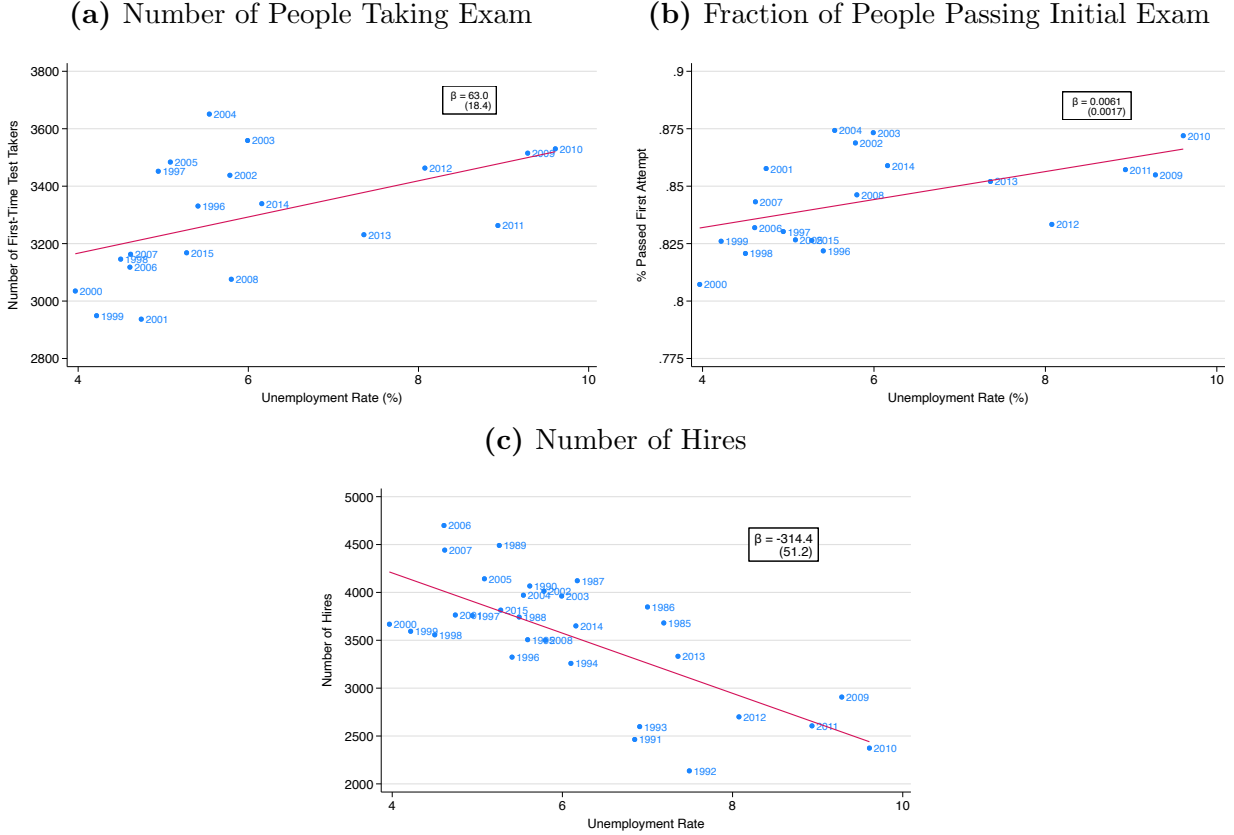
Changes Over the Cycle. Following the roadmap outlined above, we first use the full sample of SOCE test-takers to examine how the size and quality of the applicant pool vary with the business cycle. We assess the relationship between the unemployment rate in different years and the number of exam-takers, which we consider a proxy for labor supply. We present the results in the first panel of Figure 5. We find a positive correlation between the unemployment rate and the total number of exam-takers in each year, finding that a one percentage point increase in the unemployment rate leads an additional 63 candidates to take the entry-level exam.

We further examine the relationship between the unemployment rate and the fraction who pass the exam on their initial attempt, which we consider a proxy for quality of the applicant pool. Panel (b) in Figure 5 shows a positive relationship between the unemployment rate and the fraction of people who pass the exam on their initial attempt, as a one percentage point increase in the unemployment rate is associated with a 0.6 percentage point increase in the SOCE pass rate.

Passing as a Proxy for Officer Quality. In this context, we note that applicants who fail the SOCE exam are allowed to re-take the exam up to three times, and 66% of applicants who failed the test on their first try eventually pass the SOCE exam in our data. In the first column of Table 5, we show that among a sample of candidates who eventually pass the test, having passed the SOCE on the first try substantially increases the likelihood of being hired within one year of passing, from 45% to 58%. Moreover, passing the exam on the

³³Employment in industries that new officers come from – retail trade, manufacturing, construction, transportation – is more responsive to business cycle conditions than local government employment (Table A21).

Figure 5: Florida Law Enforcement: Hires and Entry Exams



Notes: The first two panels use Florida police entry exam data from 1996-2015, taken after the applicant completes recruit training. Panel (a) plots the relationship between the number of applicants who take the exam in a given year and the unemployment rate in that year. Panel (b) plots the share who pass their initial attempt and the unemployment rate in that year. Panel (c) plots the number of full-time police hires per year in Florida and the unemployment rate in each year. All panels display the linear fit, the slope coefficient, and standard error clustered at the officer's start year. Table A22 presents the regression estimates for the test-score results and documents robustness to controlling for population and a linear time trend.

first try also increases the likelihood that candidates are hired quickly (Figure A12). These results additionally suggest that officers who pass the exam on the first try have qualities that agencies value.

To further assess the link between passing on the first try and officer quality, we analyze the relationship between passing on the first try and our four outcomes of interest. We present the results in the remaining columns of Table 5, which shows that officers who passed on the first try receive 0.0043 fewer complaints than those who failed the exam on the first try, such that officers who passed receive 32% fewer complaints in their first five years of employment than their peers who failed. Furthermore, officers who passed the exam receive 0.0015 fewer disciplines, which amounts to a 40% reduction relative to their peers who did not pass on the first try. We also find significant differences in the likelihood of

Table 5: Passing the SOCE Exam on the First Try and Subsequent Outcomes

	Hired (1)	Complaints (2)	Disciplines (3)	Fired (4)	Fired for VAP (5)
Pass Exam on First Try	0.13061*** (0.00615)	-0.00428*** (0.00108)	-0.00150*** (0.00045)	-0.00951*** (0.00170)	-0.00450*** (0.00090)
Constant	0.45399*** (0.02216)	0.01015*** (0.00212)	0.00253*** (0.00074)	0.02278*** (0.00293)	0.01092*** (0.00158)
Baseline If Failed First Exam	0.4540	0.0133	0.0037	0.0242	0.0134
Current UN	—	Yes	Yes	Yes	Yes
Agency FE	—	Yes	Yes	Yes	Yes
Years Since Hire	—	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years
R^2	0.0065	0.0021	0.0005	0.0059	0.0041
Observations	61285	223578	223578	223578	223578

Notes: This table uses data from Florida police entry exams, taken after recruit training and determining hiring eligibility. Not all applicants pass the exam, and not all who pass are hired. The table examines the relationship between passing on the first try and the likelihood that officers receive complaints, disciplines, are fired, or fired for violating agency policy. We restrict the sample to officers hired in 1996-2015, the years in which we observe exam records. Column 1 estimates a specification relating the passing on the first try to the likelihood of being hired within one year of passing. Columns 2-5 use equation (1) with passing on first try as main explanatory variable. Results are robust to including the initial unemployment rate as a control. For context, we include the baseline prevalence of the five outcomes for officers who did not pass the exam on their first try. Standard errors are clustered at the officer’s starting year level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

being fired and of being fired for violating agency policy. Altogether, since passing on the first try is associated with improved measures of officer performance, these results suggest that the pool of applicants improves when outside options are scarce.

4.1.2 Predicted Complaints and Disciplines

To further understand how the quality of the applicant pool varies across the cycle, we predict the complaints and disciplines that candidates would receive if they were hired as officers (Chalfin et al., 2016, 2020). To do so, we leverage information on applicant training outcomes, applicant characteristics, and applicant exam outcomes to predict their misconduct events.

In particular, we first restrict the sample to applicants who were eventually hired as officers and use machine learning to predict whether these officers would receive complaints and/or disciplines against them based on whether they passed the SOCE on the first try, their age when they took the exam (divided into nine age brackets), and the specific academy in which they completed their training (there are 43 police training academies in Florida). Specifically, we implement a LASSO logit approach, where we select the penalization parameter through ten-fold cross-validation.³⁴

³⁴For the ten-fold cross-validation, we randomly divide the training data into ten equally-sized subsets. We estimate the model on nine subsets, and in turn validate it on the remaining subset to assess out-of-sample performance, with the process repeated ten times (Hastie et al., 2009). We randomly draw officers into the

In Table A23, we present the set of selected predictors for complaints and disciplines, respectively. Training in specific agencies predicts the likelihood of receiving complaints and disciplines, as does the age at taking the exam and having passed it on the first try. Altogether, the LASSO logit model has strong predictive power, as 6.2% of officers in the highest decile of predicted complaints actually receive a complaint against them, compared to 3.5% in the overall sample.³⁵ In turn, since we also observe the candidate predictors for applicants who were not hired, we have a measure of predicted complaints and disciplines for *all* applicants.³⁶

We subsequently correlate applicants' predicted misconduct events to the annual unemployment rate to assess how the applicant pool varies across the cycle. In Figure A13, we show that applicants in high-unemployment years have fewer predicted complaints and fewer predicted disciplines, further showing that the pool of applicants is substantively better during economic downturns.

4.2 Labor Demand

To assess the role of labor demand, in the third panel of Figure 5, we show that the number of hires in police departments in Florida is strongly correlated with the business cycle: the number of officers hired falls significantly during high unemployment years. While we do not observe information on agencies' job postings or vacancies to directly measure police agencies' labor demand, this result suggests that changes in labor demand by police departments may play a role in driving our results.

4.2.1 Characteristics of Hired Officers Across the Cycle

While we have so far shown that both the quantity and quality of applicants increases in recessions, agencies also hire fewer officers in such years, implying that labor demand may play an important role in driving our results if agencies screen out lower quality officers in recessions. For instance, agencies may adopt more stringent screening practices in high-unemployment years, thus only hiring the best applicants, such that labor demand would contribute to our main results. To assess the importance of labor demand in shaping our

training sample from all entry cohorts, ensuring that the model is not trained only on officers entering in high or low unemployment years.

³⁵Similarly, 1.7% of officers in the highest decile of predicted disciplines actually receive a discipline against them, compared to 0.9% in the overall sample. Thus, the model's lift, a measure of its predictive accuracy given by the ratio of misconduct events for officers in the top predicted decile to the average prevalence of such events, equals 1.8 and 1.9 for complaints and disciplines, respectively (Chalfin et al., 2020).

³⁶While we predict misconduct events for all applicants, the underlying model is trained only on hired officers, since outcome data on misconduct is only available for those who were actually employed as officers.

findings, we focus on the sample of applicants eligible to be hired. We analyze how the characteristics of these applicants relate to the probability of being hired and whether this relationship varies across the cycle in the following specification:

$$Hired_i = \alpha + \beta X_i + \gamma UR_i^S + \eta X_i \cdot UR_i^S + v_i \quad (2)$$

where $Hired_i$ is a binary variable capturing whether applicant i was hired within one year of passing the SOCE exam, X_i denotes a characteristic of the applicant (e.g., age, passing the exam on the initial attempt, or predicted complaints), UR_i^S captures the unemployment rate in the year they passed the SOCE exam, and η captures whether the applicants' characteristic differentially affects probability of hire across the cycle. In particular, we consider how passing the SOCE on the first try, the age at the exam, and their predicted complaints and disciplines affect the likelihood of being hired across the cycle.

We present the results in Table 6. We find that candidates who passed the exam on the first try, younger applicants and those with lower predicted complaints and disciplines are more likely to be hired in Florida within a year of passing the exam. However, the importance of these characteristics in driving the likelihood of being hired does not change across the business cycle. For instance, we do not find evidence that agencies are differentially more likely to hire first-time SOCE passers in recessions, nor are they more likely to hire applicants with low predicted complaints in such years.^{37,38} Altogether, these results indicate that while police agencies' labor demand may fall in high-unemployment years, these agencies are not changing their hiring practices across the cycle. As such, this finding further points to the importance of labor supply – through an improved applicant pool – as an important driver of our main results.³⁹

4.2.2 Bounding Exercise

To further examine the potential importance of labor demand, we assess how our results would change if agencies had not engaged in pro-cyclical hiring. Specifically, we first estimate the counterfactual hiring reduction that would have occurred in each year if the agencies in that year had faced an unemployment rate of 9.6%, the highest rate in our sample, using the

³⁷Young applicants are more likely to be hired, despite being more likely to receive complaints. However, this relationship weakens in recessions — this change in labor demand would work against our main findings.

³⁸Our results are robust to defining being hired as ever becoming an officer in Florida (Table A24). Our results are robust to using the predicted *number* of complaints/disciplines candidates would receive if hired.

³⁹To assess whether screening practices change across the cycle, we use CJAP data in 2011-2019 which includes information on agencies' use of different screening mechanisms. In Table A25, we show that agencies in Florida do not change the likelihood of using various screening mechanisms across the cycle, including requiring interviews, polygraph exams, and previous criminal justice experience, among others.

Table 6: Applicants' Characteristics and the Likelihood of Being Hired Across the Cycle

	Hired Within One Year of Passing SOCE			
	(1)	(2)	(3)	(4)
Pass Exam on First Try	0.15439*** (0.03395)			
Age at SOCE Exam		-0.01415*** (0.00264)		
Pred. Complaints			-0.03814** (0.01754)	
Pred. Disciplines				-0.01733 (0.04201)
UR at Test	-0.04384*** (0.00686)	-0.07215*** (0.00794)	-0.05054*** (0.01708)	-0.03588** (0.01265)
Variable x UR at Test	0.00003 (0.00563)	0.00103** (0.00037)	0.00208 (0.00309)	-0.00778 (0.00712)
Constant	0.69225*** (0.04855)	1.23043*** (0.05765)	0.96163*** (0.09924)	0.84185*** (0.07622)
Observations	52577	52577	52577	52577
R^2	0.030	0.035	0.024	0.024

Notes: This table estimates equation (2), assessing how applicant characteristics shape hire likelihood within 1 year of passing the SOCE exam, for candidates with observed academy training data. Column 1 considers the importance of passing the exam on the first try. Column 2 examines the role of the applicant's age at the time of the test. The last two columns consider how applicants' predicted complaints and disciplines (from Section 4.1.2) relate to the likelihood of being hired. In all columns, we examine how these characteristics affect likelihood of hire and how this effect varies across the cycle. Standard errors are clustered at the year of the exam in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

relationship between the unemployment rate and number of hires estimated in panel (c) in Figure 5.⁴⁰ We then impose that hiring reduction in the data by removing officers from the sample assuming that agencies screen out those applicants with a high predicted likelihood of police misconduct. In particular, we assume that agencies would stop hiring the officers with the highest predicted probability of having a complaint or a discipline against them in low-unemployment years.⁴¹ This is a *conservative* assumption since we show in Table 6 that agencies screening on predicted misconduct does not vary across the cycle. After 'removing' the high predicted misconduct officers from the sample, we re-estimate our main specification relating the initial unemployment rate to *actual* misconduct events, having shut down the labor demand channel. We then compare the estimated γ coefficient from this analysis and compare it to the estimated coefficients presented in Table 2 to recover the share of the main effect potentially explained by labor demand, under the conservative assumption that agencies screen more heavily on predicted misconduct in recessions.

⁴⁰For instance, in 2000 the unemployment rate is approximately 4%, which is 5.6 percentage points lower than the highest unemployment rate in our sample. We multiply 5.6 by our estimate of -314 from Figure

We present the estimated results in Figure 6. About 20-27% of the estimated effect on complaints could be explained by labor demand if agencies stopped hiring officers with high predicted complaints/disciplines in low-unemployment years. Similarly, 7-25% of the effect on disciplines could be driven by labor demand under this bounding exercise. Altogether, this analysis indicates that even when assuming that agencies screen out high-misconduct officers in low demand years, labor supply can still account for a sizable share of the overall effect.

4.2.3 Additional Tests of the Demand Channel

We further assess the importance of labor demand in our context by re-estimating our analyses in settings where labor demand forces are muted. First, we use information on agencies’ receipt of hiring grants from the Community Oriented Policing Services (COPS) program to analyze if our main results change in agencies that receive these grants (Chalfin et al., 2022).⁴² We re-estimate an interacted version of equation (1) and present our results in Table A26. We find that the estimated relationship between initial unemployment and subsequent misconduct (and firing) events does not vary by whether agencies receive hiring (and non-hiring) grants, further suggesting that labor demand does not drive our results.

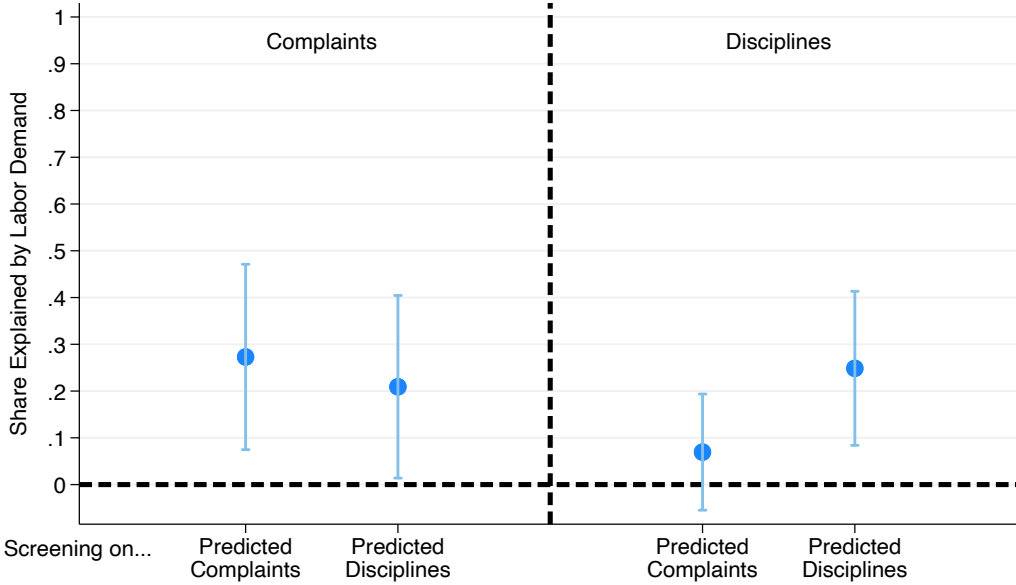
Second, we present further evidence by focusing on the relationship between initial economic conditions and subsequent officer outcomes for individuals hired in the Florida Department of Highway Safety and Motor Vehicles, which is the second largest agency in Florida. Importantly, the number of hires in this state-level agency does not fall during high unemployment periods (Figure A14). We re-estimate equation (1) for the officers hired in this department and present the results in Table A27. In this small sample of workers, we still find that officers hired in high unemployment years are significantly less likely to face disciplinary actions. Our estimate on the likelihood of being fired is similar in magnitude to the main results, albeit noisier. This result offers further suggestive evidence that our findings are primarily shaped through the labor supply mechanism posited above.

5, giving us a counterfactual hiring reduction of approximately 1,760 officers if the unemployment rate had been 9.6% in 2000.

⁴¹Agencies likely have access to additional information about applicants compared to what we observe in this context. However, for the purposes of this exercise, we note the predictive power of our machine learning model for misconduct events (as measured by the ‘area under the curve’ (AUC) in Table A23) is largely similar to that estimated by Chalfin et al. (2016) when predicting misconduct events among police officers in Philadelphia, despite the fact that they have access to extensive information on applicants’ characteristics.

⁴²If our main effects were driven by changes in labor demand across the cycle, we may expect the overall effect in Table 2 to be muted in agencies receiving COPS grants, as having access to COPS grants could allow these agencies to hire relatively more in recessions, but getting worse-quality officers. As such, if labor demand were driving the results, we would expect a weaker relationship between initial unemployment and subsequent misconduct for agencies receiving COPS grants.

Figure 6: Bounding the Effect of Labor Demand on Police Misconduct Events



Notes: This figure presents results from the bounding exercise described in Section 4.2.2. First, we use a linear regression of number of hires on the unemployment rate to estimate the hiring reduction that would have occurred in each year, if the unemployment rate had been 9.6% (the highest unemployment rate in our sample period). Then, we remove that number of officers from the data in each year. We consider two assumptions about which officers would have been screened out. Specifically, we assume that agencies rank applicants based on predicted complaints, screening out those with higher predicted complaints when labor demand falls. We then follow the same approach for predicted complaints. After dropping the officers based on predicted misconduct, we estimate our main equation on the new sample. This yields a coefficient that captures the effect of the initial unemployment rate on our outcomes after shutting down the labor demand channel. We then use that coefficient to calculate the share of the main effect that can be explained by the labor demand channel. We present that estimate of the share explained by labor demand for complaints and disciplines, under two different assumptions about agency screening. Predicted complaints and predicted disciplines are estimated through a machine learning algorithm described in Section 4.1.2. We present 95% confidence intervals based on bootstrapped standard errors from 200 replications of this procedure.

4.3 Changing Job Characteristics

As noted above, the characteristics of the job may also change during recessions and starting under certain conditions may have path-dependent effects on performance. For example, the nature of policing could vary during high unemployment periods (Horrace et al., 2022). Officers hired at different points in the business cycle may need to engage in more or less ‘proactive’ policing practices. First, since crime increases during high-unemployment periods (Raphael and Winter-Ebmer, 2001; Machin and Meghir, 2004), officers starting in those periods may need to engage in aggressive practices more often, resulting in more complaints.⁴³ Yet our main estimates show the opposite pattern. Moreover, our results are robust to directly controlling for current and starting year crime rates and to controlling for agency-specific trends in starting year (Figure A5).⁴⁴ Officers entering in high-unemployment periods also receive fewer complaints across categories that are not directly related to the policing environment, such as drug-related misconduct events (Table A30). Lastly, to account

for the possibility that the quality of veteran officers shapes the behavior of new recruits, we control for the average unemployment rate at entry of senior officers in an agency and find that our main results are robust to this control (Table A31).

At the same time, job training opportunities experienced by new officers may differ across the business cycle (Arellano-Bover, 2020). We test this using data on participation in job training courses, and we do not find a relationship between the initial unemployment rate and the likelihood of job training participation (Table A32). This suggests the quality differences we observe are not necessarily driven by differential learning opportunities across the cycle.

Overall, these results indicate that the labor supply channel likely plays a principal role in driving our main findings. Entry exam records allow us to describe the size and the quality of the applicant pool. We show, using exam outcomes and measures of predicted misconduct, that the applicant pool improves when unemployment rates are high. While agencies hire less during recessions, they do not change their hiring practices across the cycle, suggesting that the improved performance for officers hired in recessions is likely driven by changes in the applicant pool.

5 Conclusion

In this paper, we examine how economic conditions at career start affect the quality of police hires. Taking advantage of detailed data on employment spells over three decades in Florida, we show that officers hired during high-unemployment periods are of higher quality than their counterparts hired in strong labor markets, as measured by the number of complaints, disciplines, and the likelihood of being fired. We further show these effects are stronger for younger officers, who face weaker outside options during recessions. Future work should explore whether these features of the police labor market hold true in other states or in other time periods, particularly time periods or places where public confidence in police may be lower than it was in Florida from 1985-2015.

We also present novel evidence of the mechanisms underlying our findings. First, the number of people taking entry exams increases in high-unemployment years and the fraction of those who pass the initial attempt increases. Since passing the exam on the first try is associated with improved officer quality, the pool of applicants improves during high unemployment periods, indicating an important role for labor supply responses. While

⁴³In our main results, we control for the contemporaneous unemployment rate. The coefficient on that control is positive and significant (Table A28), meaning that there is a positive correlation between the current-year unemployment rate and complaints, as we argue here.

⁴⁴New hires do not enter different agencies across the cycle, where we characterize agencies by the complaints, disciplines and dismissal rates for incumbent officers (Table A29).

the number of hires is pro-cyclical in Florida, indicating a potential role for a labor demand channel, we show that even under conservative assumptions about police agencies’ screening, 15-30% of the effect of initial unemployment on officer misconduct can be explained by labor demand. As such, the evidence on labor supply and on the ability of agencies to effectively screen applicants suggests agencies could improve quality by engaging in counter-cyclical hiring. Further work should examine how agency recruitment practices can result in improved police quality, analyzing the specific aspects that agencies can change to make the occupation more appealing relative to outside options.

In general, the size of policing in the United States and the active policy debate about police reform and police funding underscores the need to understand whether financial incentives can improve policing. We make headway on this general question by leveraging variation in economic conditions and the relative appeal of outside options. While prior work offers important evidence on other public-sector occupations, policing is distinct in several ways. First, unlike teaching or nursing, policing requires relatively little formal training. Second, typical concerns about financial incentives crowding out intrinsically motivated applicants depend on positive selection into the profession, which may or may not be the case for policing. Finally, if there are fundamental issues with police culture or police training, it is possible that improving recruitment fails to change on-the-job behavior. Our findings suggest that agencies attract higher quality applicants when policing is more appealing than outside options. As a result, better officers are hired and those improvements persist even after the potentially standardizing effects of police culture and training.

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- SUPPLEMENTARY APPENDIX -
For Online Publication

A Additional Tables and Figures

Table A1: Officer Characteristics at Hire

Panel A. Characteristics of Hired Officers in Florida

	(1)
Age	29.126
Male	0.861
White	0.756
Black	0.100
Hispanic	0.123
Sample Size	100887

Panel B. Occupational Transitions of Police Officers, Current Population Survey

	New Officers Previous Occ. (1)	Old Officers Next Occ. (2)
Public Protective Service	0.219	0.195
Not Employed	0.157	0.171
Office, Administrative Support	0.101	0.102
Private Protective Service	0.095	0.113
Management	0.067	0.062
Business, Financial Operations	0.057	0.069
Sales and Related	0.041	0.033
Transportation and Material Moving	0.035	0.036
Construction and Extraction	0.032	0.028
Community, Social Service	0.031	0.033
Other	0.164	0.159
Sample Size	5,972	4,538

Notes: The first panel of Table A1 presents the age, race and gender of police officers in the FDLE data when they are first hired. The second panel takes advantage of the longitudinal component of the Current Population Survey and the Annual Social and Economic Supplement of the CPS between 1985 and 2015 to identify individuals who transition into and out of policing. We define a transition into policing as an individual who started working as a ‘Police Officers and Detectives’ and had not previously worked in that occupation or as a ‘First-Line Supervisor of Police and Detectives.’ We follow the same definition for capturing transitions out of policing. In the CPS data, we identify 5,972 individuals who become police officers at some point in the survey, and 4,538 individuals who transition out of policing. The second panel exhibits the ten most common two-digit occupations from which these individuals were previously employed in (column 1) and subsequently employed in (column 2), including those who have an unemployment spell. We split protective service occupations into public and private protective service, where the latter category includes ‘Private Detectives and Investigators,’ ‘Security Guards,’ and ‘Gaming Surveillance Officers.’

Table A2: Summary Statistics

	Within 5 Years of Hiring	All Years
Complaints	0.009 (0.100)	0.009 (0.096)
Disciplinary Actions	0.003 (0.053)	0.002 (0.049)
Fired	0.015 (0.122)	0.011 (0.103)
Fired for Violating Agency Policy	0.009 (0.096)	0.007 (0.084)
Voluntary Separation	0.083 (0.276)	0.063 (0.244)
Unemployment Rate in Starting Year	5.929 (1.355)	5.828 (1.187)
Unemployment Rate in Current Year	6.019 (1.471)	6.116 (1.582)
Observations	468,924	941,593

Notes: This table produces summary statistics based on each year of employment for the sample of full time police officers beginning an employment spell from 1985-2015. Column 1 limits to the first 0-5 years of an officer's employment spell while column 2 includes all years. The rows are defined as follows: number of complaints per year, number of disciplinary actions per year, likelihood of being fired in a given year, likelihood of being fired for violating agency policy in a given year, likelihood of separating voluntarily in a given year, the starting year unemployment rate for officers in the sample, and the current year unemployment rate.

Table A3: Robustness to Controlling for Annual Discipline Rate

	Main Specification				Control for Annual Discipline Rate			
	Complaints (1)	Disciplines (2)	Fired (3)	Fired VAP (4)	Complaints (5)	Disciplines (6)	Fired (7)	Fired VAP (8)
UR in Starting Year	-0.00077*** (0.00027)	-0.00039*** (0.00013)	-0.00073** (0.00027)	-0.00044** (0.00017)	-0.00098*** (0.00024)	-0.00056*** (0.00010)	-0.00087*** (0.00027)	-0.00078*** (0.00017)
Constant	0.01089*** (0.00154)	0.00314*** (0.00058)	0.01808*** (0.00247)	0.00973*** (0.00160)	0.00697*** (0.00220)	0.00002 (0.00081)	0.01544*** (0.00263)	0.00374** (0.00140)
Observations	468921	468921	468921	468921	468921	468921	468921	468921
Agency FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Years Since Hire	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years
R ²	0.0028	0.0007	0.0043	0.0034	0.0029	0.0009	0.0044	0.0037

Notes: This table tests the robustness of our main findings to controlling for an agency's contemporaneous disciplinary environment. Columns 1-4 replicate our main specification from Table 2 for each of the four outcomes. Columns 5-8 augment this specification by adding a control for the annual discipline rate. The analysis is restricted to the first five years of an officer's career. All specifications include agency fixed effects and a control for the current year unemployment rate. Standard errors, clustered at the officer's starting year, are in parentheses. * p<0.1, ** p<0.05, *** p<0.01.

Table A4: Effect of Starting Year Unemployment on Officer Decertification

	Decertification	
	(1)	(2)
Unemployment Rate in Starting Year	-0.0022 (0.0016)	-0.0021 (0.0015)
Constant	0.0393*** (0.0094)	0.0391*** (0.0090)
Agency FE	No	Yes
R ²	0.0003	0.0044
Observations	75209	75209

Notes: This table estimates the relationship between the unemployment rate at career start and the likelihood that an officer is decertified from policing in Florida. The second column includes agency-at-hire fixed effects. Note, we cannot estimate our typical specification for this outcome since date of decertification is missing for a large share of officers. Standard errors are clustered at the officer's starting year level in parentheses. * p<0.1, ** p<0.05, *** p<0.01.

Table A5: Effect of Starting Year Unemployment on Officer Quality,
Within-Florida Unemployment Rates

Panel A: Five-Year Window

	Complaints (1)	Disciplines (2)	Fired (3)	Fired for VAP (4)
Unemployment Rate in Starting Year	-0.00017 (0.00018)	-0.00010* (0.00005)	-0.00048*** (0.00016)	-0.00044*** (0.00014)
Constant	0.01072*** (0.00105)	0.00315*** (0.00030)	0.01811*** (0.00092)	0.01158*** (0.00081)
Agency-Year FE	Yes	Yes	Yes	Yes
Years Since Hire	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years
R^2	0.0163	0.0050	0.0148	0.0155
Observations	382952	382952	382952	382952

Panel B: All Years

	Complaints (1)	Disciplines (2)	Fired (3)	Fired for VAP (4)
Unemployment Rate in Starting Year	-0.00017** (0.00007)	-0.00009*** (0.00003)	-0.00025* (0.00012)	-0.00022*** (0.00007)
Constant	0.00992*** (0.00043)	0.00270*** (0.00015)	0.01273*** (0.00068)	0.00851*** (0.00045)
Agency-Year FE	Yes	Yes	Yes	Yes
Years Since Hire	All Years	All Years	All Years	All Years
R^2	0.0132	0.0061	0.0173	0.0144
Observations	706564	706564	706564	706564

Notes: In this table, we take advantage of data on county-level unemployment rates — which we aggregate up to seven sub-state regions, including Fort Myers, Jacksonville, Miami, Orlando, Pensacola, Tallahassee and Tampa Bay — to assess the effect of local economic conditions on officer quality. We estimate the following specification leveraging within-state variation in unemployment rates at career start on officer quality:

$$Y_{iat} = \alpha + \gamma UR_{ir}^S + \theta_{at} + \varepsilon_{iat} \quad (\text{A1})$$

where UR_{ir}^S captures the unemployment rate in region r faced by officer i at career start; θ_{at} captures agency-by-year fixed effects. This analysis does not incorporate potential within-state migration responses to local economic conditions. Moreover, local unemployment rates are only available starting in 1990. Panel A estimates equation (A1) restricting the analysis to the first five years of an officer's employment spell, whereas Panel B includes all years of an officer's employment spell. See Table 2 notes for a description of the outcome variables. Standard errors are clustered at the starting year by region level in parentheses.

* p<0.1, ** p<0.05, *** p<0.01

Table A6: Local Variation in Incidence of the Great Recession: Pre- and Post-Recession Hiring Outcomes

	Complaints		Disciplines		Fired		Fired VAP	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post $\times$ Severe	-0.0056* (0.0029)	-0.0047 (0.0053)	-0.0009 (0.0014)	-0.0007 (0.0022)	-0.0058 (0.0049)	-0.0057 (0.0070)	-0.0056 (0.0033)	-0.0077* (0.0045)
Constant	0.0234*** (0.0005)	0.0233*** (0.0008)	0.0054*** (0.0002)	0.0053*** (0.0003)	0.0515*** (0.0008)	0.0515*** (0.0011)	0.0225*** (0.0005)	0.0228*** (0.0007)
Observations	20723	20710	20723	20710	20723	20710	20723	20710
Start Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Local FE	CZ	Agency	CZ	Agency	CZ	Agency	CZ	Agency

Notes: This table presents results from a spatial analysis examining whether pre- and post-Great Recession differences in officer quality vary by recession severity across Florida commuting zones. Each column reports results from the following equation: $Y_{ict} = \alpha_c + \gamma_t + \lambda(\text{Post}_t \times \text{Severe}_c) + \epsilon_{ict}$, where the dependent variable is measured during officers' first two years of employment. The coefficient of interest is the interaction between *Post* (indicator for officers hired 2009–2011) and *Severe* (indicator for commuting zones experiencing above-median recession shocks within Florida). Columns 1–2 report complaints; 3–4 disciplinary actions; 5–6 fired; 7–8 fired for violating agency policy (VAP). Within each outcome, odd-numbered columns include commuting-zone fixed effects and even-numbered columns include agency fixed effects. All specifications include start-year fixed effects. Standard errors are clustered at the commuting-zone or agency level as indicated. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A7: Agency-Level Employment Composition and Crime Related Outcomes

	(Log) Crimes		Clearance Rates		(Log) Violent Crimes		(Log) Other Crimes	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Agency's Mean UR in Starting Year	-0.0432*	-0.0627**	0.0169	0.0179*	-0.0615	-0.0492	-0.0388	-0.0605**
	(0.0254)	(0.0248)	(0.0108)	(0.0096)	(0.0469)	(0.0393)	(0.0284)	(0.0263)
County Unemployment	0.0095	0.0145**	-0.0028	-0.0040	0.0033	0.0051	0.0116*	0.0168***
	(0.0061)	(0.0061)	(0.0041)	(0.0066)	(0.0089)	(0.0080)	(0.0063)	(0.0062)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Agency FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Agency Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Population	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Local Unemployment	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Linear Trend	No	Yes	No	Yes	No	Yes	No	Yes
R^2	0.9740	0.9807	0.1387	0.1593	0.9500	0.9612	0.9704	0.9776
Observations	7006	7006	6849	6849	6968	6968	7006	7006

Notes: This table presents evidence on the relationship between agencies' employment composition and their crime and clearance rates. We match agency-level annual outcomes from the Unified Crime Report (UCR) in 1985-2015, covering information on the number of crimes committed in the area assigned to each agency and the number of cases cleared. We relate these outcomes to agency-level characteristics in the following specification:

$$Y_{at} = \alpha + \gamma \overline{UR_{at}^S} + \overline{X_{at}} + \eta X_{ct} + \lambda_t + \theta_a + e_{at} \quad (\text{A2})$$

where Y_{at} captures one of the crime-related outcomes described above. $\overline{UR_{at}^S}$ represents the average unemployment rate at career start among all officers employed in agency a in year t . $\overline{X_{at}}$ captures agency-level characteristics, including the agencies' race, gender and age composition as well as the number of officers employed in a particular year. X_{ct} includes county-level time-varying characteristics, including county-level unemployment and its population. The first two columns use log crimes as the outcome variable of interest. Columns (3) and (4) use the annual agency-level crime clearance rate as the outcome of interest, defined as the number of clearances divided by the number of crimes in that year. The next two columns use log violent crimes as the main outcome, whereas columns (7) and (8) focus on (log) other types of crimes. The odd-numbered columns control for agency and year fixed effects, for average agency-characteristics, as well as for population and local unemployment rates. Even-numbered columns add an agency-specific linear trend. Standard errors are clustered at the agency level in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Starting Year Unemployment and Officer Quality,
Excluding 1985-94

Panel A: Five-Year Window

	Complaints	Disciplinary Actions	Fired	Fired for VAP
	(1)	(2)	(3)	(4)
Unemployment Rate in Starting Year	-0.00069** (0.00026)	-0.00042*** (0.00009)	-0.00045** (0.00019)	-0.00043*** (0.00015)
Constant	0.0115*** (0.0014)	0.0034*** (0.0005)	0.0167*** (0.0021)	0.0091*** (0.0010)
P-value: $\beta_{robust}=\beta_{baseline}$	0.6121	0.8520	0.2446	0.7853
Observations	318,075	318,075	318,075	318,075
R^2	0.0035	0.0010	0.0051	0.0036

Panel B: Two-Year Window

	Complaints	Disciplinary Actions	Fired	Fired for VAP
	(1)	(2)	(3)	(4)
Unemployment Rate in Starting Year	-0.00113*** (0.00034)	-0.00068*** (0.00015)	-0.00098*** (0.00031)	-0.00072** (0.00029)
Constant	0.0102*** (0.0011)	0.0027*** (0.0003)	0.0187*** (0.0022)	0.0094*** (0.0011)
P-value: $\beta_{robust}=\beta_{baseline}$	0.7633	0.7732	0.1920	0.7302
Observations	194,497	194,497	194,497	194,497
R^2	0.0038	0.0012	0.0059	0.0041

Notes: This table presents estimates of the effect of the starting year unemployment rate on officer quality, excluding the starting years 1985-1994. Each column reports results from a separate regression following our main specification (equation 1). All specifications include agency fixed effects and control for the current unemployment rate. The sample consists of officers between 1995 and 2015. Panel A restricts the analysis to the first five years of an officer's employment spell, whereas Panel B restricts the analysis to the first two years. Standard errors clustered at the officer's starting year level are shown in parentheses. The row "P-value: $\beta_{robust}=\beta_{baseline}$ " reports the p-value from a statistical test of equality between our baseline estimates and the estimates displayed in this table. * p<0.1, ** p<0.05, *** p<0.01

Table A9: Effect of Starting Year Unemployment on Officer Quality,
Monthly Unemployment

Panel A. Concurrent Unemployment at Start

	Complaints (1)	Disciplinary Actions (2)	Fired (3)	Fired for VAP (4)
Unemployment Rate in Starting Month	-0.00004** (0.00002)	-0.00002** (0.00001)	-0.00007*** (0.00002)	-0.00004*** (0.00001)
Constant	0.0006*** (0.0001)	0.0001*** (0.0000)	0.0017*** (0.0003)	0.0010*** (0.0001)
Observations	4508353	4508353	4508353	4508353
R^2	0.0003	0.0001	0.0005	0.0004

Panel B. Lagged Unemployment at Start (6- and 12-Months)

	Complaints (1)	Disciplinary Actions (2)	Fired (3)	Fired for VAP (4)	Complaints (5)	Disciplinary Actions (6)	Fired (7)	Fired for VAP (8)
Lagged UN Rate	-0.00006** (0.00002)	-0.00002*** (0.00001)	-0.00007** (0.00002)	-0.00004** (0.00001)	-0.00007*** (0.00002)	-0.00002*** (0.00001)	-0.00006** (0.00003)	-0.00003* (0.00002)
Constant	0.0007*** (0.0001)	0.0002*** (0.0000)	0.0017*** (0.0003)	0.0010*** (0.0001)	0.0008*** (0.0001)	0.0002*** (0.0000)	0.0017*** (0.0003)	0.0010*** (0.0001)
UR Start Lag	6 Months				12 Months			
Observations	4508353	4508353	4508353	4508353	4508353	4508353	4508353	4508353
R^2	0.0003	0.0001	0.0005	0.0004	0.0003	0.0001	0.0005	0.0004

Notes: Table A9 presents results from equation (1) using monthly unemployment rates and month-level outcomes for each officer. The first panel presents estimates of the relationship between the unemployment rate in the officer's starting month and the prevalence of the four quality measures considered in the paper (see notes for Table 2). The second panel examines the relationship between lagged unemployment rates at start and month-level outcomes. The first four columns present evidence using a six-month lag in the initial unemployment rate. The last four columns present evidence using a twelve-month lag in the initial unemployment rate. Note, the coefficients in this table are much smaller than the coefficients in our main specification because the outcomes are at the monthly level. Multiplying the coefficients in this table by 12 yields estimates that are more comparable to our year-level results. Standard errors are clustered at the officer's starting year level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A10: Coefficient Estimates from Leave-Two-Out Exercise

First Year of Excluded Pair	Complaints	Disciplines	Fired	Fired VAP	Mean % Change from Baseline Estimates
1985	-0.00060	-0.00040	-0.00040	-0.00036	-21.12
1986	-0.00067	-0.00040	-0.00060	-0.00046	-5.57
1987	-0.00074	-0.00037	-0.00074	-0.00044	-1.76
1988	-0.00072	-0.00030	-0.00071	-0.00035	-12.37
1989	-0.00071	-0.00031	-0.00068	-0.00033	-15.25
1990	-0.00082	-0.00039	-0.00074	-0.00043	2.29
1991	-0.00094	-0.00048	-0.00082	-0.00054	20.30
1992	-0.00086	-0.00044	-0.00077	-0.00050	10.64
1993	-0.00076	-0.00038	-0.00071	-0.00045	-0.65
1994	-0.00081	-0.00039	-0.00072	-0.00045	2.05
1995	-0.00081	-0.00039	-0.00072	-0.00044	1.47
1996	-0.00075	-0.00038	-0.00075	-0.00044	0.31
1997	-0.00070	-0.00039	-0.00082	-0.00048	3.40
1998	-0.00073	-0.00040	-0.00083	-0.00051	7.53
1999	-0.00075	-0.00038	-0.00080	-0.00051	5.47
2000	-0.00073	-0.00038	-0.00081	-0.00052	5.93
2001	-0.00075	-0.00039	-0.00076	-0.00049	3.48
2002	-0.00080	-0.00040	-0.00072	-0.00046	2.21
2003	-0.00078	-0.00040	-0.00073	-0.00045	1.48
2004	-0.00076	-0.00041	-0.00082	-0.00049	7.23
2005	-0.00084	-0.00046	-0.00089	-0.00050	16.02
2006	-0.00072	-0.00051	-0.00075	-0.00039	4.57
2007	-0.00064	-0.00045	-0.00066	-0.00041	-4.01
2008	-0.00084	-0.00038	-0.00067	-0.00038	-3.51
2009	-0.00106	-0.00038	-0.00078	-0.00032	4.29
2010	-0.00099	-0.00033	-0.00089	-0.00046	9.85
2011	-0.00074	-0.00027	-0.00078	-0.00046	-5.39
2012	-0.00063	-0.00030	-0.00068	-0.00038	-14.92
2013	-0.00067	-0.00034	-0.00069	-0.00039	-10.91
2014	-0.00077	-0.00039	-0.00073	-0.00046	1.10

Notes: This table presents estimates of the effect of the starting year unemployment rate on officer quality, excluding all consecutive two-year pairs starting with 1985-86 and ending with 2014-15. Each row and column reports results from a separate regression following our main specification (equation 1), but excluding the pair of years beginning with the year listed in the first column. All specifications include agency fixed effects and control for the current unemployment rate. The sample consists of officers in their first five years of employment. The final column reports the mean percentage change, based on the estimates from cols. 1-4, relative to our baseline estimates. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A11: Starting Year Unemployment and Officer Quality, Alternative Clustering

Panel A: Agency Clustering

	Complaints	Disciplinary Actions	Fired	Fired for VAP
	(1)	(2)	(3)	(4)
Unemployment Rate in Starting Year	-0.00077*** (0.00014)	-0.00039*** (0.00006)	-0.00073*** (0.00015)	-0.00044*** (0.00012)
Constant	0.0109*** (0.0011)	0.0031*** (0.0004)	0.0181*** (0.0012)	0.0097*** (0.0008)
Observations	468,924	468,924	468,924	468,924
R^2	0.0028	0.0007	0.0043	0.0034

Panel B: Person Clustering

	Complaints	Disciplinary Actions	Fired	Fired for VAP
	(1)	(2)	(3)	(4)
Unemployment Rate in Starting Year	-0.00077*** (0.00012)	-0.00039*** (0.00006)	-0.00073*** (0.00013)	-0.00044*** (0.00011)
Constant	0.0109*** (0.0008)	0.0031*** (0.0004)	0.0181*** (0.0010)	0.0097*** (0.0007)
Observations	468,924	468,924	468,924	468,924
R^2	0.0028	0.0007	0.0043	0.0034

Notes: See notes for Table 2. This table reproduces those main results, but clusters at the agency-level in the first panel and at the officer-level in the second panel. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A12: Effect of Starting Year Unemployment on Officer Quality,
Wild Cluster Bootstrap

	Complaints	Disciplinary Actions	Fired	Fired for VAP
	(1)	(2)	(3)	(4)
Unemployment Rate in Starting Year	-0.00077*** (0.00027)	-0.00039*** (0.00013)	-0.00073** (0.00027)	-0.00044** (0.00017)
Constant	0.0109*** (0.0015)	0.0031*** (0.0006)	0.0181*** (0.0025)	0.0097*** (0.0016)
Wild Cluster Bootstrap 95% CI	[-0.0013,-0.0002]	[-0.0007,-0.0001]	[-0.0013,-0.0002]	[-0.0008,-0.0001]
Wild Cluster Bootstrap P-Value	0.015	0.004	0.004	0.007
Observations	468,924	468,924	468,924	468,924
R^2	0.0028	0.0007	0.0043	0.0034

Notes: See notes for Table 2. This table reproduces those main results, but adds the 95 percent confidence interval and the p-value derived from a wild cluster bootstrap procedure. * p<0.1, ** p<0.05, *** p<0.01

Table A13: Robustness to First-Time Officers Only

	Main Sample				First-Time Officers Only			
	Complaints (1)	Disciplines (2)	Fired (3)	Fired for VAP (4)	Complaints (5)	Disciplines (6)	Fired (7)	Fired for VAP (8)
UR in Starting Year	-0.00077*** (0.00027)	-0.00039*** (0.00013)	-0.00073** (0.00027)	-0.00044** (0.00017)	-0.00073** (0.00033)	-0.00036** (0.00014)	-0.00076** (0.00033)	-0.00049** (0.00020)
Constant	0.01089*** (0.00154)	0.00314*** (0.00058)	0.01808*** (0.00247)	0.00973*** (0.00160)	0.00951*** (0.00183)	0.00300*** (0.00069)	0.01961*** (0.00309)	0.01010*** (0.00169)
Observations	468921	468921	468921	468921	328422	328422	328422	328422
Agency FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Years Since Hire	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years
R^2	0.0028	0.0007	0.0043	0.0034	0.0014	0.0002	0.0040	0.0032

Notes: This table tests the robustness of our findings to restricting the sample to first-time police officers. Columns 1-4 replicate our main specification from Table 2, which includes all officer employment spells. Columns 5-8 re-estimate the same specification on a restricted sample that includes only an officer's first observed employment spell in the FDLE data. The analysis is restricted to the first five years of each employment spell. All specifications include agency fixed effects and a control for the current year unemployment rate. Standard errors, clustered at the officer's starting year, are presented in parentheses. * p<0.1, ** p<0.05, *** p<0.01.

Table A14: Effect of Starting Year Unemployment on Officer Quality by Age

Panel A: Linear Age Interaction

	Complaints (1)	Disciplines (2)	Fired (3)	Fired for VAP (4)
Unemployment Rate in Starting Year	-0.00229*** (0.00082)	-0.00117** (0.00047)	-0.00413*** (0.00099)	-0.00233*** (0.00073)
Age at Start	-0.00053*** (0.00011)	-0.00027*** (0.00006)	-0.00123*** (0.00016)	-0.00075*** (0.00011)
Age $\times$ UR Start	0.00004** (0.00002)	0.00002** (0.00001)	0.00009*** (0.00002)	0.00005*** (0.00002)
Constant	0.03036*** (0.00491)	0.01299*** (0.00267)	0.06345*** (0.00609)	0.03753*** (0.00545)
Current UN	Yes	Yes	Yes	Yes
Agency FE	Yes	Yes	Yes	Yes
Years Since Hire	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years
R^2	0.0035	0.0013	0.0071	0.0053
Observations	467447	467447	467447	467447

Panel B: Young (Age 30) Interaction

	Complaints (1)	Disciplines (2)	Fired (3)	Fired for VAP (4)
Unemployment Rate in Starting Year	-0.00064** (0.00025)	-0.00033*** (0.00009)	-0.00040 (0.00025)	-0.00031** (0.00014)
Young	0.00888*** (0.00263)	0.00489*** (0.00153)	0.02514*** (0.00381)	0.01490*** (0.00308)
Young $\times$ UR Start	-0.00072* (0.00042)	-0.00038 (0.00023)	-0.00197*** (0.00056)	-0.00105** (0.00043)
Constant	0.00880*** (0.00153)	0.00201*** (0.00048)	0.01224*** (0.00267)	0.00640*** (0.00126)
Current UN	Yes	Yes	Yes	Yes
Agency FE	Yes	Yes	Yes	Yes
Years Since Hire	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years
R^2	0.0033	0.0012	0.0068	0.0051
Observations	468921	468921	468921	468921

Notes: See notes for Table 4. In Panel A, instead of using a binary term to capture officer age, the results in this table are based on a model that uses a linear term in age. In Panel B, we include a binary variable that equals one if the officer is below 30 upon starting (i.e. young) and equal to zero if not, and an interaction between that variable and the unemployment rate. Standard errors are clustered at the officer's starting year level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A15: Effect of Starting Year Unemployment on Officer Quality by Race and Gender

Panel A: Heterogeneity by Race				
	Complaints (1)	Disciplines (2)	Fired (3)	Fired for VAP (4)
Unemployment Rate in Starting Year	-0.00071** (0.00032)	-0.00040 (0.00025)	-0.00080** (0.00033)	-0.00017 (0.00030)
White	-0.00375*** (0.00128)	-0.00182 (0.00125)	-0.00923*** (0.00209)	-0.00159 (0.00192)
White $\times$ UR Start	-0.00008 (0.00019)	0.00002 (0.00019)	0.00009 (0.00034)	-0.00038 (0.00031)
Constant	0.01387*** (0.00212)	0.00456*** (0.00130)	0.02530*** (0.00268)	0.01110*** (0.00252)
Current UN	Yes	Yes	Yes	Yes
Agency FE	Yes	Yes	Yes	Yes
Years Since Hire	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years
R^2	0.0031	0.0008	0.0051	0.0037
Observations	468921	468921	468921	468921

Panel B: Heterogeneity by Gender				
	Complaints (1)	Disciplines (2)	Fired (3)	Fired for VAP (4)
Unemployment Rate in Starting Year	-0.00068** (0.00027)	-0.00019 (0.00011)	-0.00079* (0.00046)	-0.00026 (0.00031)
Male	0.00385* (0.00195)	0.00309*** (0.00080)	-0.00150 (0.00306)	0.00297 (0.00217)
Male $\times$ UR Start	-0.00011 (0.00033)	-0.00023* (0.00013)	0.00008 (0.00044)	-0.00022 (0.00034)
Constant	0.00761*** (0.00180)	0.00050 (0.00067)	0.01935*** (0.00324)	0.00719*** (0.00208)
Current UN	Yes	Yes	Yes	Yes
Agency FE	Yes	Yes	Yes	Yes
Years Since Hire	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years
R^2	0.0029	0.0008	0.0043	0.0034
Observations	468921	468921	468921	468921

Notes: See notes for Table 4. Instead of exploring heterogeneity in effects by officer age, this table explores heterogeneity by other officer demographics. Panel A tests for heterogeneity in effect size by race and Panel B tests for heterogeneity in effect size by gender. Standard errors are clustered at the officer's starting year level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A16: Effect of Starting Year Unemployment Rate on Officer Quality
by Agency Unionization Status

	Complaints (1)	Disciplines (2)	Fired (3)	Fired for VAP (4)
Unemployment Rate in Starting Year	-0.00070** (0.00029)	-0.00036** (0.00017)	-0.00077* (0.00038)	-0.00054* (0.00032)
Agency Has Union $\times$ UR Start	-0.00014 (0.00029)	-0.00003 (0.00017)	0.00015 (0.00035)	0.00016 (0.00028)
Constant	0.01199*** (0.00167)	0.00339*** (0.00067)	0.01804*** (0.00240)	0.01000*** (0.00176)
Current UN	Yes	Yes	Yes	Yes
Agency FE	Yes	Yes	Yes	Yes
Years Since Hire	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years
R^2	0.0023	0.0005	0.0038	0.0030
Observations	394385	394385	394385	394385

Notes: This table displays results estimated from equation (1) but including a binary variable equal to one if the officer joined an agency that was unionized by the time of entry and equal to zero if not, along with an interaction between that variable and the starting year unemployment rate. Standard errors clustered at the level of officer's starting year in parentheses. Each observation corresponds to an officer-by-year employment record. Columns 1-4 are estimated on the same outcomes described in the notes for Table 2. All columns limit the sample to years 0-5 from each officer's starting year. All specifications include a control for current year unemployment rate and agency fixed effects. Table A2 reports the overall mean of each of these outcomes within the first five years of an officer's career. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A17: Heterogeneity by Agency Size

Panel A: Inverse Agency-Size Weighting

	Complaints (1)	Disciplines (2)	Fired (3)	Fired for VAP (4)
Unemployment Rate in Starting Year	-0.00146*** (0.00038)	-0.00063*** (0.00016)	-0.00138*** (0.00050)	-0.00105*** (0.00029)
Constant	0.01827*** (0.00266)	0.00417*** (0.00080)	0.02273*** (0.00393)	0.01501*** (0.00260)
Current UN	Yes	Yes	Yes	Yes
Agency FE	Yes	Yes	Yes	Yes
Years Since Hire	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years
R^2	0.0114	0.0045	0.0138	0.0110
Observations	468921	468921	468921	468921

Panel B. Heterogeneity Across Large Agencies

	Complaints (1)	Disciplines (2)	Fired (3)	Fired for VAP (4)
Bottom 3 Quartiles × UR Start	-0.00133*** (0.00034)	-0.00064*** (0.00016)	-0.00098** (0.00047)	-0.00077** (0.00030)
Top Quartile Agency Size × UR Start	-0.00057** (0.00028)	-0.00030** (0.00013)	-0.00064** (0.00027)	-0.00033* (0.00017)
Constant	0.01081*** (0.00154)	0.00310*** (0.00058)	0.01804*** (0.00247)	0.00968*** (0.00161)
Current UN	Yes	Yes	Yes	Yes
Agency FE	Yes	Yes	Yes	Yes
Years Since Hire	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years
R^2	0.0028	0.0007	0.0043	0.0034
Observations	468921	468921	468921	468921

Notes: This table examines heterogeneity in the effect of the starting year unemployment rate on officer quality by agency size. Panel A reports results from regressions where each observation is weighted by the inverse of the number of officers in the agency's hiring-year cohort. This approach assigns equal weight to each agency-year, down-weighting observations from larger agencies. Panel B reports results from a specification that interacts the starting year unemployment rate with an indicator for being hired into a large agency. Large agencies are defined as those with more than 80 officers, corresponding to the 75th percentile of the agency size distribution. The sample is restricted to the first five years of an officer's career. All specifications include agency fixed effects and a control for the current year unemployment rate. Standard errors are clustered at the officer's starting year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A18: Effect of Starting Year Unemployment Rate on Hired Officer Demographics

	White (1)	Black (2)	Hispanic (3)	Male (4)	Age (5)	White (6)	Black (7)	Hispanic (8)	Male (9)	Age (10)
Unemployment Rate in Starting Year	0.00065 (0.00642)	-0.00062 (0.00126)	-0.00067 (0.00473)	0.00020 (0.00124)	-0.17782 (0.10704)	-0.00232 (0.00606)	-0.00006 (0.00109)	0.00174 (0.00461)	-0.00016 (0.00126)	-0.20931* (0.12114)
Constant	0.73779*** (0.03814)	0.10717*** (0.00733)	0.13901*** (0.02794)	0.85535*** (0.00767)	36.11348*** (0.67098)	0.75538*** (0.03640)	0.10384*** (0.00624)	0.12473*** (0.02727)	0.85746*** (0.00783)	36.30045*** (0.75175)
Agency FE	No	No	No	No	No	Yes	Yes	Yes	Yes	Yes
R^2	0.0000	0.0000	0.0000	0.0000	0.0006	0.1975	0.0694	0.1880	0.0122	0.0900
Observations	103971	103971	103971	103971	103664	103971	103971	103971	103971	103664

Notes: This table estimates the relationship between unemployment rate in a given year and the demographic characteristics of officers hired in that year. Columns 1-3 and 6-8 test whether economic conditions influence the race or ethnicity of new hires. Columns 4 and 9 test whether economic conditions affect the sex of new hires. Finally, columns 5 and 10 test whether economic conditions impact the age of new hires. Columns 1-5 exclude agency fixed effects and columns 6-10 include them. Standard errors are clustered at the year level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A19: SOCE Performance and Differential Attrition

	Voluntary Separation		Fired	
	(1)	(2)	(3)	(4)
UR Start	0.008 (0.006)	0.008 (0.006)	-0.003 (0.004)	-0.003 (0.004)
Pass SOCE on First Try	-0.015 (0.037)	0.008 (0.035)	-0.050** (0.021)	-0.041* (0.021)
Variable x UR at Start	-0.009 (0.007)	-0.008 (0.006)	0.003 (0.004)	0.002 (0.004)
Constant	0.371*** (0.036)	0.347*** (0.034)	0.114*** (0.021)	0.110*** (0.021)
Agency FE	No	Yes	No	Yes
R^2	0.001	0.128	0.001	0.030
Observations	33164	33164	33164	33164

Notes: This table tests for differential attrition by officer quality across the business cycle. The unit of observation is an officer. The dependent variables are indicators for separating voluntarily in the first five years (first two columns) and fired within five years (last two columns). The key test is the coefficient on the interaction term ‘Variable x UR at Start,’ which interacts a measure of officer quality (passing the SOCE on the first try) with the unemployment rate at career start. The estimated coefficients on the interaction terms suggest that high- and low-quality officers do not separate at different rates depending on the economic conditions at hire. Even-numbered columns include agency fixed effects. Standard errors are clustered at the officer’s starting year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A20: Unemployment Rate and Starting Salaries in Policing

	Log Starting Wages		
	(1)	(2)	(3)
County-Level Unemployment	0.0036 (0.0040)	0.0064 (0.0054)	0.0008 (0.0048)
Constant	10.7041*** (0.0241)	10.7281*** (0.0323)	10.6818*** (0.0296)
Agencies	All	Police Dept.	Sheriff's Office
Year FE	Yes	Yes	Yes
Agency FE	Yes	Yes	Yes
R^2	0.8787	0.8202	0.9461
Observations	2741	2155	586

Notes: we use Florida Criminal Justice Agency Profile (CJAP) data in 2011-2019 to examine how posted entry salaries in police agencies and sheriff's offices vary across the business cycle. We estimate the following specification: $W_{at}^S = \alpha + \beta UR_{ct} + \lambda_t + \theta_a + v_{at}$, where W_{at}^S denotes either a police agency or sheriff's office posted entry salary in a given year over the period 2011-2019. We cluster standard errors at the county-year level, presented in parentheses. * p<0.1, ** p<0.05, *** p<0.01

Table A21: Business Cycle and Industry-Level Employment

	Private (1)	Construction (2)	Manufacturing (3)	Retail Trade (4)	Transportation (5)	Professional (6)	Local Govt. (7)
Unemployment Rate	-0.021*** (0.003)	-0.053*** (0.008)	-0.039*** (0.003)	-0.017*** (0.003)	-0.030*** (0.003)	-0.028*** (0.007)	-0.001 (0.004)
Constant	11.344*** (0.021)	8.718*** (0.054)	10.286*** (0.025)	9.463*** (0.021)	8.077*** (0.024)	9.062*** (0.049)	8.340*** (0.029)
R^2	0.959	0.757	0.968	0.873	0.975	0.948	0.890
Observations	35	35	35	35	35	35	35

Notes: In this table, we use Bureau of Labor Statistics data on industry-level annual employment outcomes to assess the cyclical of employment across different sectors in the following specification: $E_{st} = \alpha + \gamma UR_t + t + \varepsilon_{st}$ where E_t represents the natural logarithm of total employment in industry s in year t , UR_t denotes the national unemployment rate in year t and t represents a linear time trend. We focus on the years 1985-2019 to match the time period considered in the paper. The outcome variable in the first column encompasses total private sector non-farm employment. The second through sixth columns consider employment in the construction, manufacturing, retail trade, transportation and professional and business services sectors, respectively. The last column focuses on local government employment excluding education, which serves as a proxy for police employment since disaggregated data on total police employment is not directly available. Robust standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01

Table A22: Unemployment Rate and Test-Taking Outcomes

	Number of Test-Takers (1)	Share of Passed Exams (2)	Number of Passed Exams (3)
Unemployment Rate	89.3484*** (20.1710)	0.0070*** (0.0021)	82.9233*** (22.8315)
Constant	5.2167 (2142.7524)	0.4460** (0.1904)	-2751.6923 (2583.2535)
FL Population	Yes	Yes	Yes
Linear Time Trend	Yes	Yes	Yes
R^2	0.2425	0.2767	0.2660
Observations	20	20	20

Notes: This table uses data from Florida police entry exams, taken after the applicant completes basic recruit training. These data are available from 1996-2015. The first column examines the number of annual test-takers as the outcome of interest. The second column presents evidence on the share of passed exams. The last column analyzed the total number of passed exams in each year. All specifications include the Florida population as a control variable and a linear time trend as a control. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A23: Machine Learning Predictors for Complaints and Disciplines

	Complaints (1)	Disciplines (2)
Pass Exam on First Try	Yes	Yes
Age		
18 to 19	Yes	
45 to 49	Yes	Yes
50 to 54	Yes	Yes
Police Academy		
Citrus County Public Safety Training Center	Yes	Yes
Daytona State College, School Of Emergency Services	Yes	
Eastern Florida State College, Public Safety Institute	Yes	
Florida Panhandle Technical College Public Safety Institute	Yes	
Florida Keys Community College		Yes
George Stone Area Vo-Tech Center		Yes
Indian River State College		Yes
Manatee Technical College	Yes	
Miami Police Training Center	Yes	Yes
Northeast Florida Criminal Justice Center	Yes	
Pat Thomas Law Enforcement Academy		Yes
Polk State College-K. C. Thompson Institute Of Public Safety	Yes	Yes
Santa Fe College Institute Of Public Safety	Yes	
Southwest Florida Public Service Academy	Yes	
Suncoast Technical College		Yes
Observations	42032	42032
AUC	0.5517	0.5716

Notes: This table presents the selected predictors from the machine learning exercise described in Section 4.1.2. The first two columns present the selected predictors for the analysis using the sample of police officer applicants for their predicted complaints and disciplines, respectively, from the set of available predictors that include passing on the first try, nine categories of age at the SOCE exam, and their police academy agency. The last row includes the ‘area under the curve’ (AUC) for each estimated model, where the AUC assesses the model’s accuracy in predicting misconduct events.

Table A24: Applicants' Characteristics and the Likelihood of Being Hired Across the Cycle

	Hired As a Police Officer in Florida			
	(1)	(2)	(3)	(4)
Pass Exam on First Try	0.11786*** (0.02720)			
Age at SOCE Exam		-0.01371*** (0.00324)		
Pred. Complaints			-0.01988** (0.00852)	
Pred. Disciplines				0.02071 (0.02978)
UR at Test	-0.02942*** (0.00523)	-0.04796*** (0.01022)	-0.03280*** (0.00834)	-0.02046** (0.00821)
Variable x UR at Test	-0.00151 (0.00428)	0.00062 (0.00046)	0.00070 (0.00129)	-0.01070** (0.00479)
Constant	0.76540*** (0.03439)	1.26084*** (0.07852)	0.93850*** (0.05556)	0.84900*** (0.05135)
Observations	52577	52577	52577	52577
R^2	0.017	0.035	0.013	0.014

Notes: This table displays results estimated from equation (2). We assess how applicant characteristics shape the differential likelihood of ever getting hired in policing in Florida across the cycle for a sample of candidates for whom we observe information on their police academy training outcomes. The first column considers the importance of passing the SOCE exam on the first try, the second column examines the role of the applicants' age at the time of the test. The last two columns consider how applicants' predicted complaints and disciplines (from the machine learning exercise described in Section 4.1.2) relate to the likelihood of being hired in policing in Florida. Clustered standard errors at the year of exam presented in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A25: Relationship between Unemployment Rate and Screening Practices

	Education	Interview	Polygraph	Psych. Exam	Fitness	Agility Test	CJ Experience	Swimming	Driving
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
County-Level UR	0.0209 (0.0200)	-0.0002 (0.0085)	-0.0093 (0.0096)	0.0101 (0.0107)	-0.0175 (0.0166)	0.0040 (0.0147)	-0.0018 (0.0083)	-0.0002 (0.0054)	-0.0018 (0.0129)
Constant	12.1287*** (0.1254)	0.9377*** (0.0528)	0.5871*** (0.0590)	0.7414*** (0.0669)	0.6837*** (0.1044)	0.5453*** (0.0924)	0.0850* (0.0513)	0.0948*** (0.0341)	0.8648*** (0.0807)
Agencies	All	All	All	All	All	All	All	All	All
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Agency FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.7019	0.4921	0.8577	0.7849	0.6106	0.7352	0.6097	0.8486	0.5206
Observations	2840	2842	2842	2841	2842	2841	2841	2842	2842

Notes: We use Florida Criminal Justice Agency Profile (CJAP) data in 2011-2019 to examine how screening practices in police agencies and sheriff's offices in Florida vary across the business cycle. We estimate the following specification: $Screening_{at} = \alpha + \beta UR_{ct} + \lambda_t + \theta_a + v_{at}$, where $Screening_{at}$ denotes a screening practice adopted by police agencies and sheriff's offices in Florida in a given year over the period 2011-2019. The table presents evidence, in order across columns, for whether agencies are more likely to require that applicants have a minimum number of years of completed schooling, to require them to do an interview in front of a board, to take a polygraph exam, to undergo a psychological examination, to complete a fitness program, to complete a physical fitness agility test, to have previous experience in criminal justice, to take a swimming test, and to share their driving history prior to being hired. Across all columns, we fail to find evidence that the prevalence of screening requirements changes across the cycle. We cluster standard errors at the county-year level, presented in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A26: Effect of Starting Year Unemployment Rate on Officer Quality,
Heterogeneity by COPS Grant Receipt

	Complaints	Disciplinary Actions	Fired	Fired for VAP
	(1)	(2)	(3)	(4)
Unemployment Rate in Starting Year	-0.00080*** (0.00028)	-0.00040*** (0.00014)	-0.00072** (0.00028)	-0.00043** (0.00017)
UR x Hiring Grants in Starting Year	0.00000 (0.00001)	0.00000 (0.00000)	0.00000 (0.00001)	-0.00000 (0.00000)
UR x Non-hiring Grants in Starting Year	0.00005 (0.00004)	0.00002 (0.00003)	0.00001 (0.00004)	-0.00002 (0.00005)
Constant	0.0111*** (0.0016)	0.0032*** (0.0006)	0.0181*** (0.0026)	0.0096*** (0.0016)
Observations	468,924	468,924	468,924	468,924
R^2	0.0028	0.0006	0.0043	0.0034

Notes: This table displays results estimated from equation 1, with an interaction term for the amount of funding the agency receives in hiring and in non-hiring grants in each year. In particular, we use information on agencies' receipt of hiring grants from the Community Oriented Policing Services (COPS) program to analyze if our main results change in agencies that receive these grants. Standard errors are clustered at the officer's starting year level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A27: Effect of Starting Year Unemployment Rate on Officer Quality, Department of Highway Patrol

	Complaints	Disciplinary Actions	Fired	Fired for VAP
	(1)	(2)	(3)	(4)
Unemployment Rate in Starting Year	-0.00027 (0.00137)	-0.00072*** (0.00023)	-0.00099 (0.00083)	-0.00042 (0.00033)
Constant	-0.0069 (0.0061)	0.0014 (0.0015)	0.0159 (0.0095)	0.0003 (0.0030)
Observations	16,044	16,044	16,044	16,044
R^2	0.0016	0.0007	0.0000	0.0005

Notes: See Table 2 notes for a description of the outcome variables. This table estimates equation (1) restricting the sample to officers employed in the Florida Department of Highway Safety and Motor Vehicles, which is the second largest agency in Florida. We limit to this agency because as a state-wide agency its hiring is not as responsive to business cycle conditions (Figure A14). We still find that officers hired in high unemployment years are significantly less likely to face disciplinary actions. Our estimate on the likelihood of being fired is similar in magnitude to the main results, but it is noisier. Standard errors are clustered at the officer's starting year level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A28: Effect of Starting Year Unemployment Rate on Officer Quality, Contemporaneous Unemployment

	Complaints	Disciplinary Actions	Fired	Fired for VAP
	(1)	(2)	(3)	(4)
Unemployment Rate in Starting Year	-0.00077*** (0.00027)	-0.00039*** (0.00013)	-0.00073** (0.00027)	-0.00044** (0.00017)
Contemporaneous Unemployment Rate	0.00050* (0.00026)	0.00032*** (0.00011)	0.00021 (0.00034)	0.00037** (0.00017)
Constant	0.0109*** (0.0015)	0.0031*** (0.0006)	0.0181*** (0.0025)	0.0097*** (0.0016)
Observations	468,924	468,924	468,924	468,924
R^2	0.0028	0.0007	0.0043	0.0034

Notes: This table displays results estimated from equation (1), including the estimated coefficient on the contemporaneous unemployment rate. The coefficient on the initial unemployment rate is also presented in Table 2. Each observation corresponds to an officer-by-year employment record. See Table 2 notes for a description of the outcome variables. Standard errors are clustered at the officer's starting year level in parentheses in all specifications. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A29: Effect of Starting Year Unemployment on Agency Characteristics

	Complaints (1)	Disciplines (2)	Fired (3)	Fired for VAP (4)
Unemployment Rate in Starting Year	-0.00007 (0.00005)	-0.00001 (0.00001)	-0.00003 (0.00009)	-0.00003 (0.00005)
Constant	0.00152*** (0.00030)	0.00049*** (0.00008)	0.00253*** (0.00052)	0.00192*** (0.00029)
R^2	0.0002	0.0000	0.0000	0.0000
Observations	106134	106134	106134	106134

Notes: This table displays the relationship between new officers' unemployment rate at hire and the average characteristics of the agency they entered, where agencies are represented by their (residualized) incumbent officers' complaints, disciplines, proportion fired and proportion fired for violating agency policy across the four columns. The residualized characteristics are computed as follows. First, for each agency-year in our data, we first compute the average values of each of these characteristics only for incumbent officers. We then residualize these averages by regressing each characteristic against a sub-state region fixed effect and incumbent-officer-level characteristics, including their age, race and gender. We then relate the residualized characteristics to the national-level unemployment rate for new officers. Standard errors are clustered at the level of the officer's starting year in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A30: Effect of Starting Year Unemployment on Officer Complaint Types

	All Complaints (1)	Violent (2)	Drug-Related (3)	Other (4)
Unemployment Rate in Starting Year	-0.00077*** (0.00027)	-0.00026*** (0.00008)	-0.00004** (0.00002)	-0.00045** (0.00021)
Constant	0.01089*** (0.00154)	0.00344*** (0.00057)	0.00056*** (0.00014)	0.00689*** (0.00114)
Baseline Prevalence	0.0094	0.0025	0.0004	0.0065
Current UN	Yes	Yes	Yes	Yes
Agency FE	Yes	Yes	Yes	Yes
Years Since Hire	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years
R^2	0.0028	0.0009	0.0001	0.0023
Observations	468921	468921	468921	468921

Notes: This table estimates the equation (1) using different complaint types as the outcome variable. The first column replicates the result presented in the first column of Table 2. The second column analyzes the prevalence of violent complaints, which are due to arson, assault, homicide, kidnapping, robbery and/or sex offenses. The third column examines drug-related complaints, which include those due to selling, possessing, smuggling, trafficking, or distributing drugs such as hallucinogens, heroin, cocaine, and marijuana, among others. Lastly, the 'Other' complaints comprise misconduct events due to bribery, burglary, commercialized sexual offenses, damage property, family offenses, forgery, fraud, invasion of privacy, larceny, obscenity, obstructing the judiciary, Congress, legislature or the police, violating public peace, sexual assault, smuggling, stealing property, traffic offenses, and weapon offenses. Standard errors are clustered at the officer's starting year level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A31: Robustness to Controlling for Veterans' Average UR at Start

	Main Specification				Control for Veterans' UR at Start			
	Complaints (1)	Disciplines (2)	Fired (3)	Fired VAP (4)	Complaints (5)	Disciplines (6)	Fired (7)	Fired VAP (8)
UR in Starting Year	-0.00077*** (0.00027)	-0.00039*** (0.00013)	-0.00073** (0.00027)	-0.00044** (0.00017)	-0.00070*** (0.00024)	-0.00043*** (0.00012)	-0.00064*** (0.00022)	-0.00054*** (0.00016)
Constant	0.01089*** (0.00154)	0.00314*** (0.00058)	0.01808*** (0.00247)	0.00973*** (0.00160)	0.01739*** (0.00390)	-0.00146 (0.00205)	0.02648*** (0.00741)	-0.00065 (0.00516)
Observations	468921	468921	468921	468921	468921	468921	468921	468921
Agency FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Years Since Hire	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years	≤ 5 Years
R ²	0.0028	0.0007	0.0043	0.0034	0.0028	0.0007	0.0044	0.0035

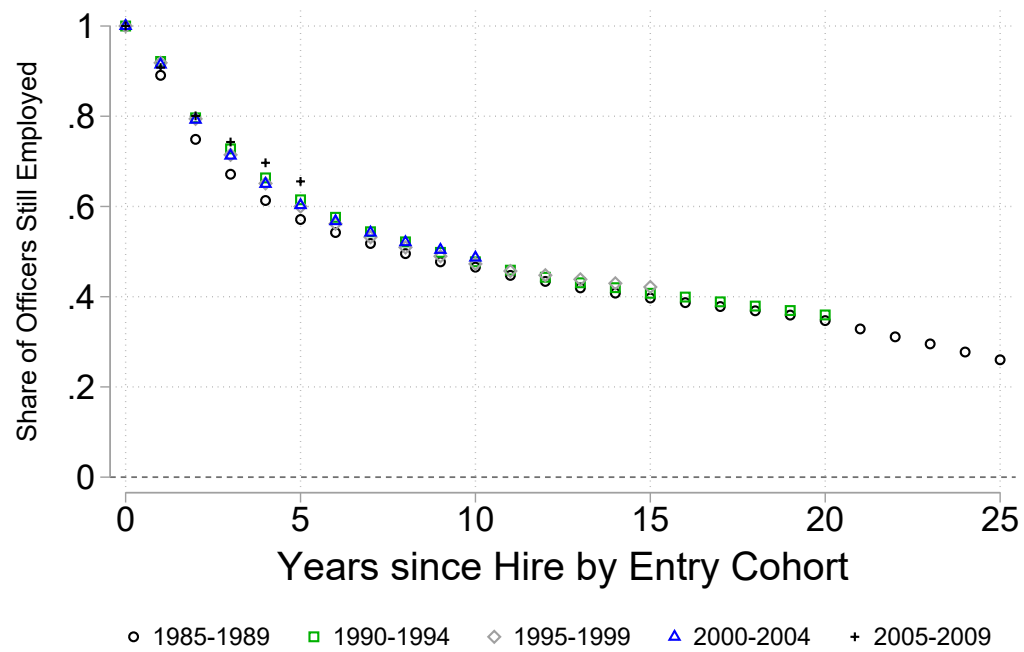
Notes: This table tests the robustness of our main findings to controlling for veteran officers' average unemployment rate at entry. Columns 5-8 augment this specification with a control for the average starting unemployment rate of veteran officers. This control is defined as the average starting unemployment rate of all veteran officers (i.e., those hired in a previous year) employed in an officer's agency at the time of their hire. The analysis is restricted to the first five years of an officer's career. All specifications include agency fixed effects and a control for the current year unemployment rate. Standard errors, clustered at the officer's starting year, are in parentheses. * p<0.1, ** p<0.05, *** p<0.01.

Table A32: Effect of Starting Year Unemployment on Job Training Participation

	Training Courses		Hours Trained	
	Advanced (1)	Specialized (2)	Advanced (3)	Specialized (4)
Unemployment Rate	-0.0012 (0.0031)	-0.0023 (0.0026)	-0.0845 (0.1436)	-0.0062 (0.0497)
Constant	0.3682*** (0.0366)	0.0295 (0.0232)	16.5538*** (1.6536)	0.2983 (0.3755)
Observations	468921	468921	468921	468921

Notes: This table displays results estimated from equation (1) on officers' job training participation. The first two columns present the number of courses that officers participated in and passed in each year. The last two columns show the number of hours of job training participation in each year, where the number of hours are given by the minimum number of hours required for each course. Advanced courses include 117 different courses, including courses in radar speed measurement, interviews and interrogations, narcotics and dangerous drugs investigation, line supervision, stress management techniques, and advanced report writing and review, among others. Specialized courses include breath test operator courses, general instructor refresher course, laser speed measuring device courses and defensive tactics and firearms instruction, among others. Standard errors clustered at the level of officer's starting year in parentheses. Each observation corresponds to an officer-by-year employment record. * p<0.1, ** p<0.05, *** p<0.01

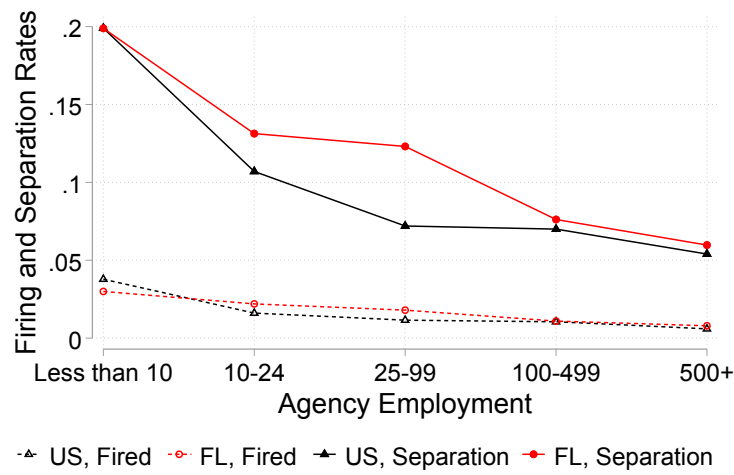
Figure A1: Proportion Employed by Years since Hire



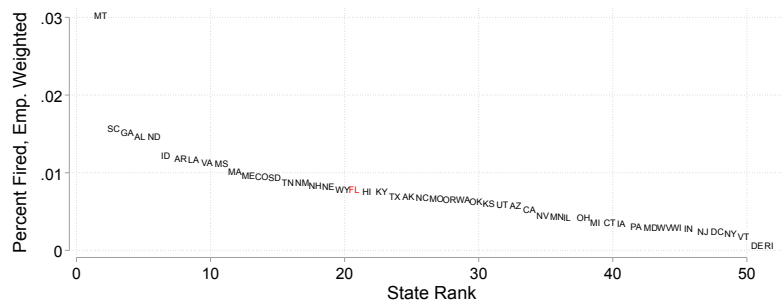
Notes: This figure shows the share of officers who remained employed in each year since their initial hire. We present evidence for five different cohorts of entrants, encompassing those who entered in 1985-1989, 1990-1994, 1995-1999, 2000-2004 and 2005-2009.

Figure A2: Labor Market Dynamics for Police Officers, Florida and United States

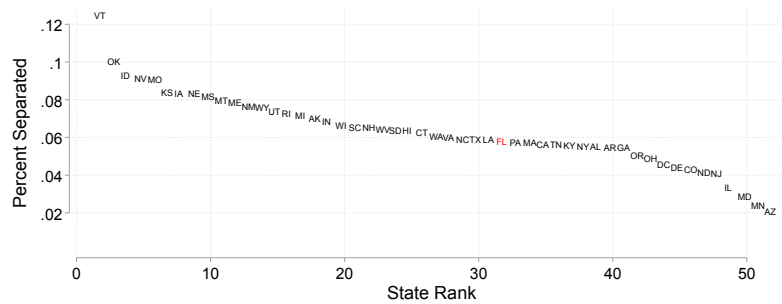
(a) Firing and Separation Rates by Agency Size, BJS 2008



(b) Firing Rates by State, LEMAS 2003

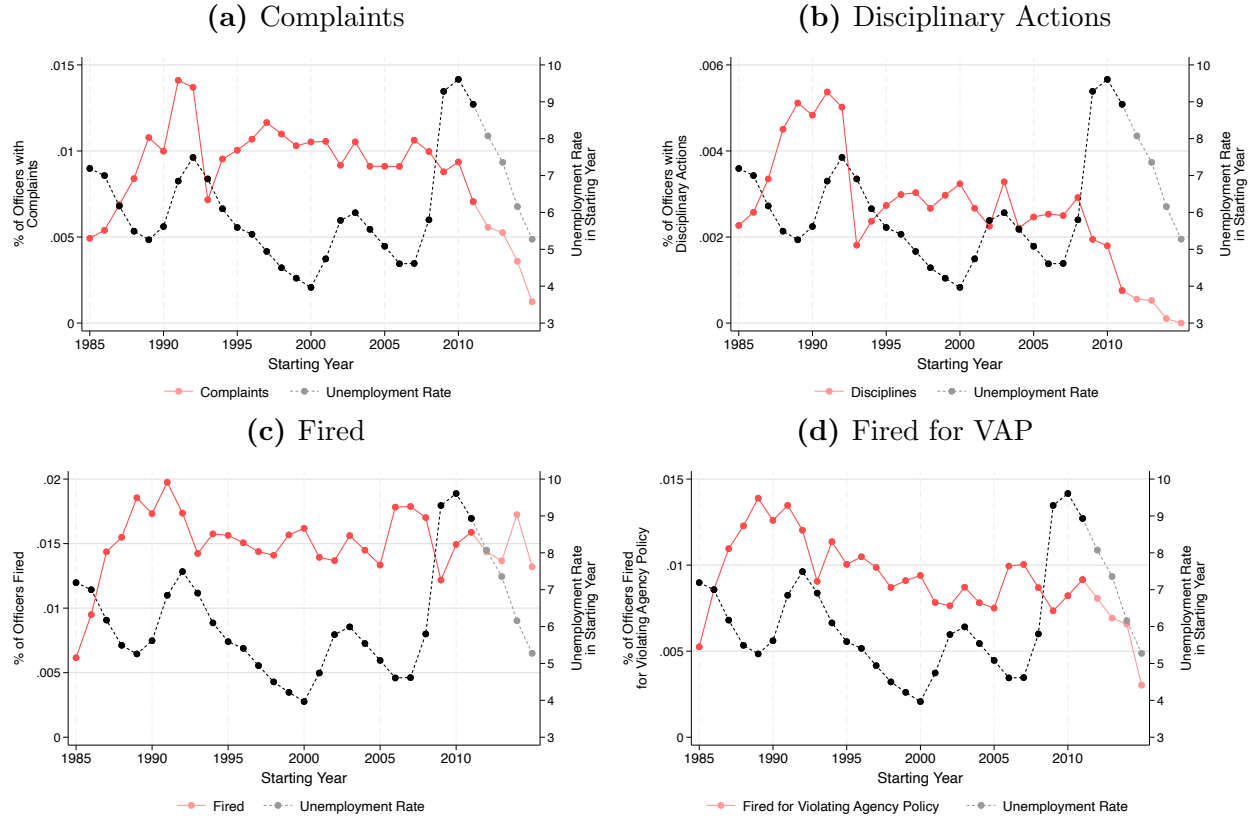


(c) Separation Rates by State, CPS 1985-2015



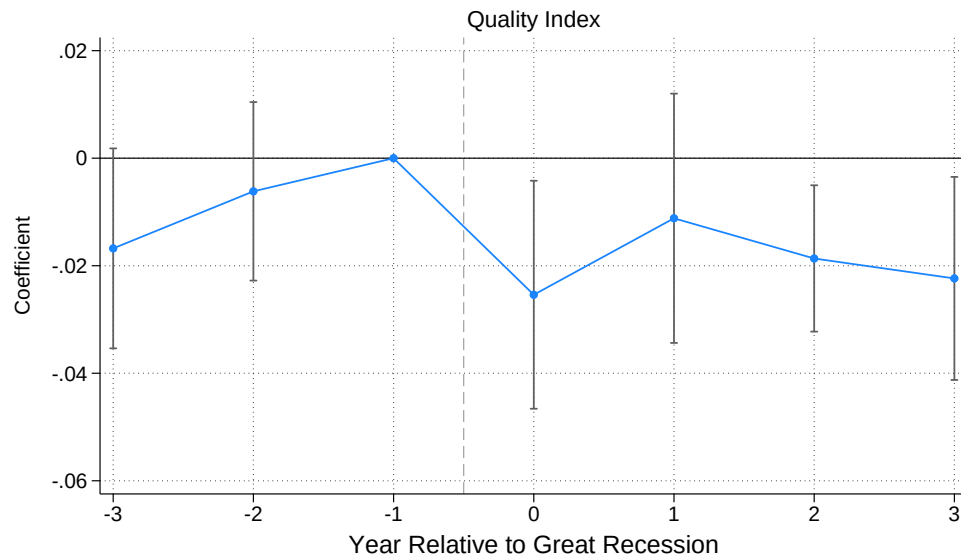
Notes: The first panel compares firing and separation rates by agency size in Florida to a nationwide sample from a 2008 report by the Bureau of Justice Statistics. The second panel compares firing and separation rates in Florida to all other states using a nationwide sample from the 2003 LEMAS. The last panel compares separation rates in Florida to all other states using information from the Current Population Survey across 1985-2015.

Figure A3: Relationship Between Starting Year Unemployment Rate and Officer Outcomes



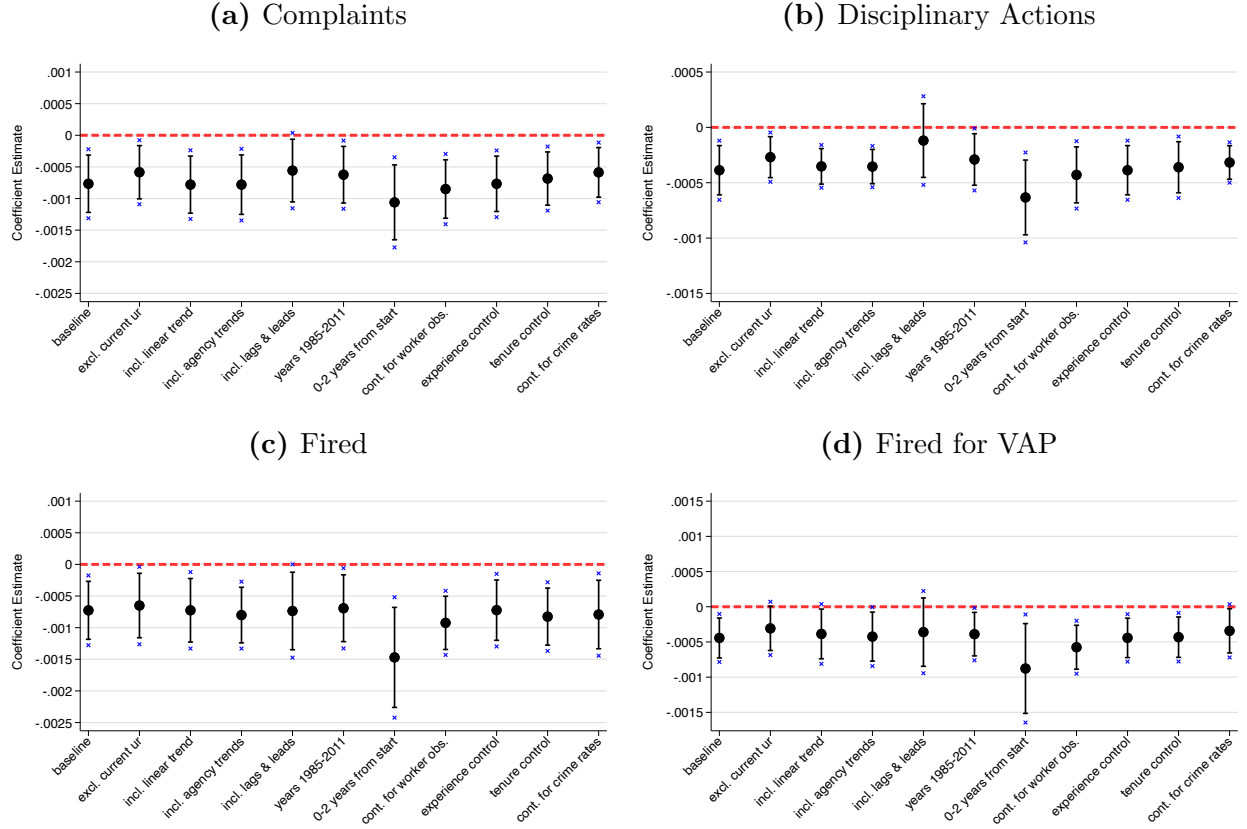
Notes: This figure plots the relationship between the unemployment rate at career start and the four officer quality outcomes in our analysis. Each panel shows the average annual rate of the outcome for officers by their starting year cohort (red solid line, left y-axis) and the national unemployment rate in the starting year (blue dashed line, right y-axis). The correlation coefficient between each outcome and the unemployment rate is reported at the top of each panel. For each entry cohort, we calculate the average annual rate of each outcome within the first five years of employment. Complaints and disciplinary actions are measured as the average number of events per officer per year, while firing outcomes are measured as the average probability of being fired per year (in percentage points). Although we report raw means in this figure, the time-series for these outcomes are virtually identical ($\text{corr} = 0.99$) if we instead estimate them after accounting for agency fixed effects. The figure uses data from all officers who began their employment between 1985 and 2015, with outcomes observed through 2016.

Figure A4: Local Variation Incidence of the Great Recession: Pre- and Post-Recession Hiring Outcomes, Event Study



Notes: This figure presents event-study estimates comparing the evolution of officer quality in severely versus mildly affected commuting zones around the Great Recession, following [Yagan \(2019\)](#) and [Weinstein \(2025\)](#). Commuting zones are classified as severely or mildly affected based on whether their 2007–2009 change in the local unemployment rate falls above or below the Florida median. The outcome is equal to one if the officer has any complaint or is fired within their first two years of employment and zero otherwise. Each point plots the year-by-year differential in the outcome between these two groups, with the coefficient normalized to zero in 2007. We include commuting zone fixed effects and start-year fixed effects. Standard errors are clustered at the commuting-zone level and 95 percent confidence intervals are presented.

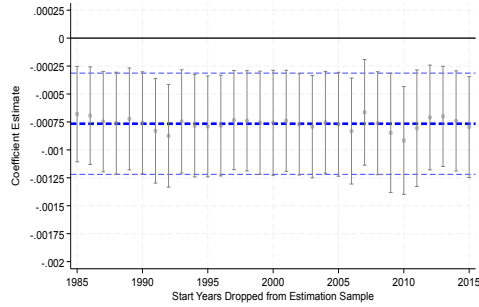
Figure A5: Effect of Starting Year Unemployment Rate on Officer Outcomes, Robustness Tests



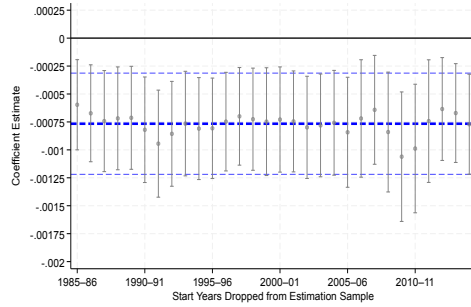
Notes: This figure shows an array of robustness tests that we conducted for our main results. In the following order, it shows: our baseline estimate from Table 2, excluding the current year unemployment rate, including a linear time trend, including agency-specific linear trends in the starting year, including leads and lags of the starting year unemployment rate, restricting to years 1985-2011, restricting to the first 0-2 years of an officer's career, controlling for worker observable characteristics, controlling for their experience as officers, controlling for officers' tenure within their agency, and controlling for national current and starting year crime rates (results are robust to using Florida crime rates). With each coefficient, we plot the 90 percent confidence interval, based on standard errors clustered at the level of officer's starting year. The blue dots include the 95% confidence interval for each estimate. See Table 2 notes for a description of the outcome variables.

Figure A6: Effect of Starting Year Unemployment Rate on Officer Outcomes, Dropping Entry Cohorts

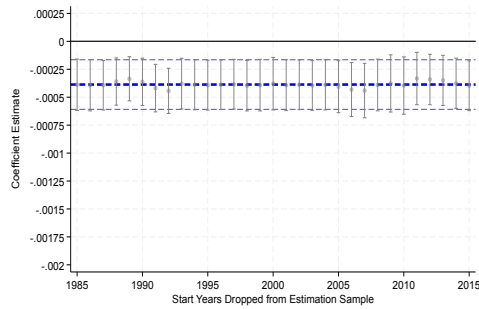
(a) Complaints: Drop One Cohort



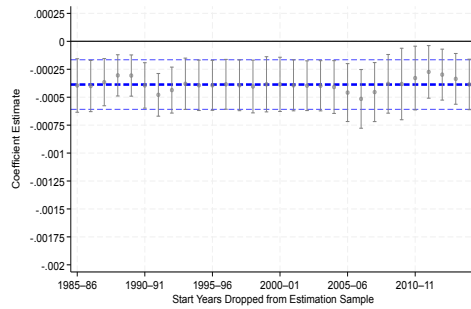
(b) Complaints: Drop Two Cohorts



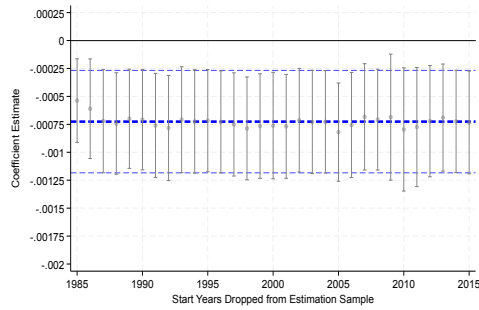
(c) Disciplines: Drop One Cohort



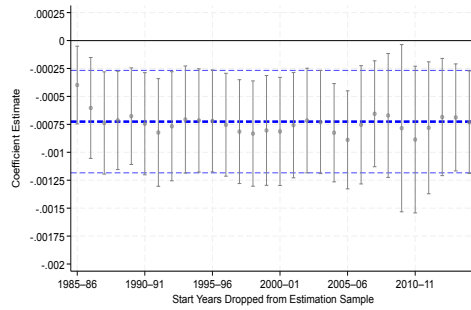
(d) Disciplines: Drop Two Cohorts



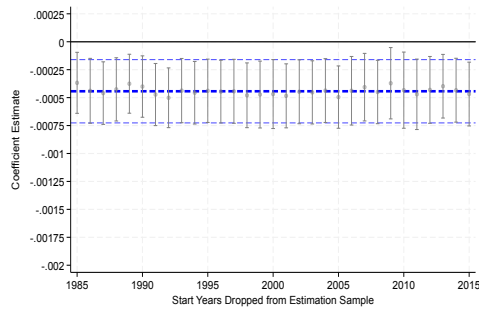
(e) Fired: Drop One Cohort



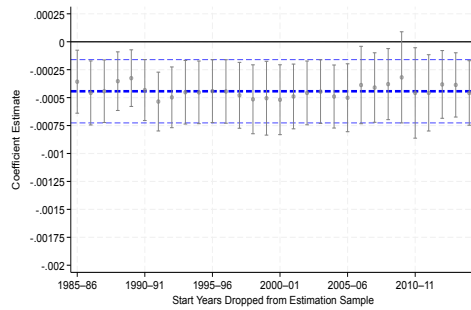
(f) Fired: Drop Two Cohorts



(g) Fired VAP: Drop One Cohort



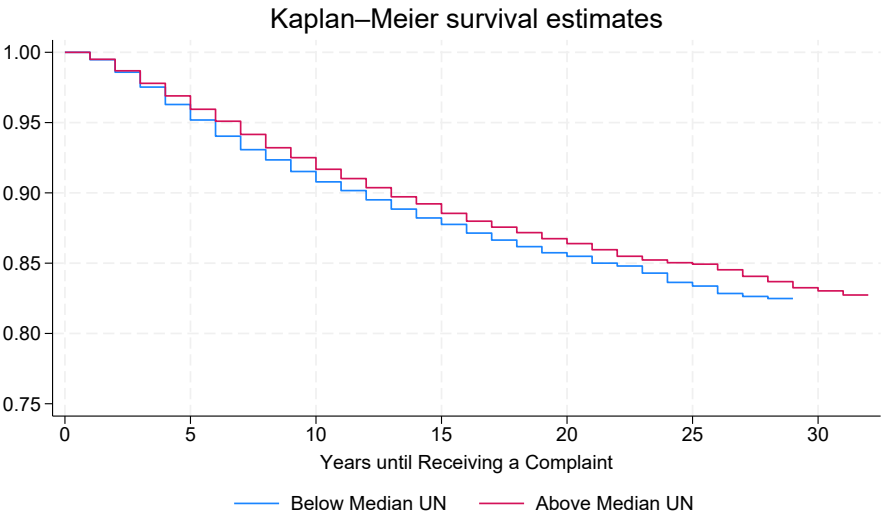
(h) Fired VAP: Drop Two Cohorts



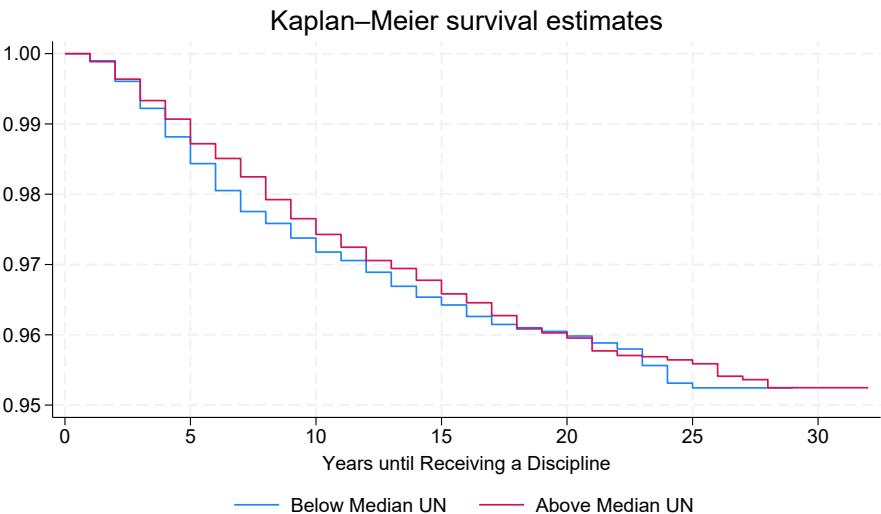
Notes: This figure shows the robustness of our estimates to dropping specific entry cohorts: dropping one and two cohorts at a time for complaints (panels (a) and (b), respectively); disciplines (panels (c) and (d)), being fired (panels (e) and (f)), and being fired for violating agency policy (panels (g) and (h)). Each point represents the estimated coefficient from our main specification (Equation 1 in the paper) after dropping officers who started in an entry year and across two adjacent entry years. The shaded area represents the 90% confidence interval for each estimate, with standard errors clustered at the officer's starting year level. See Table 2 notes for a description of the outcome variables.

Figure A7: Survival Function Until First Misconduct Event by Initial Unemployment

(a) Complaints

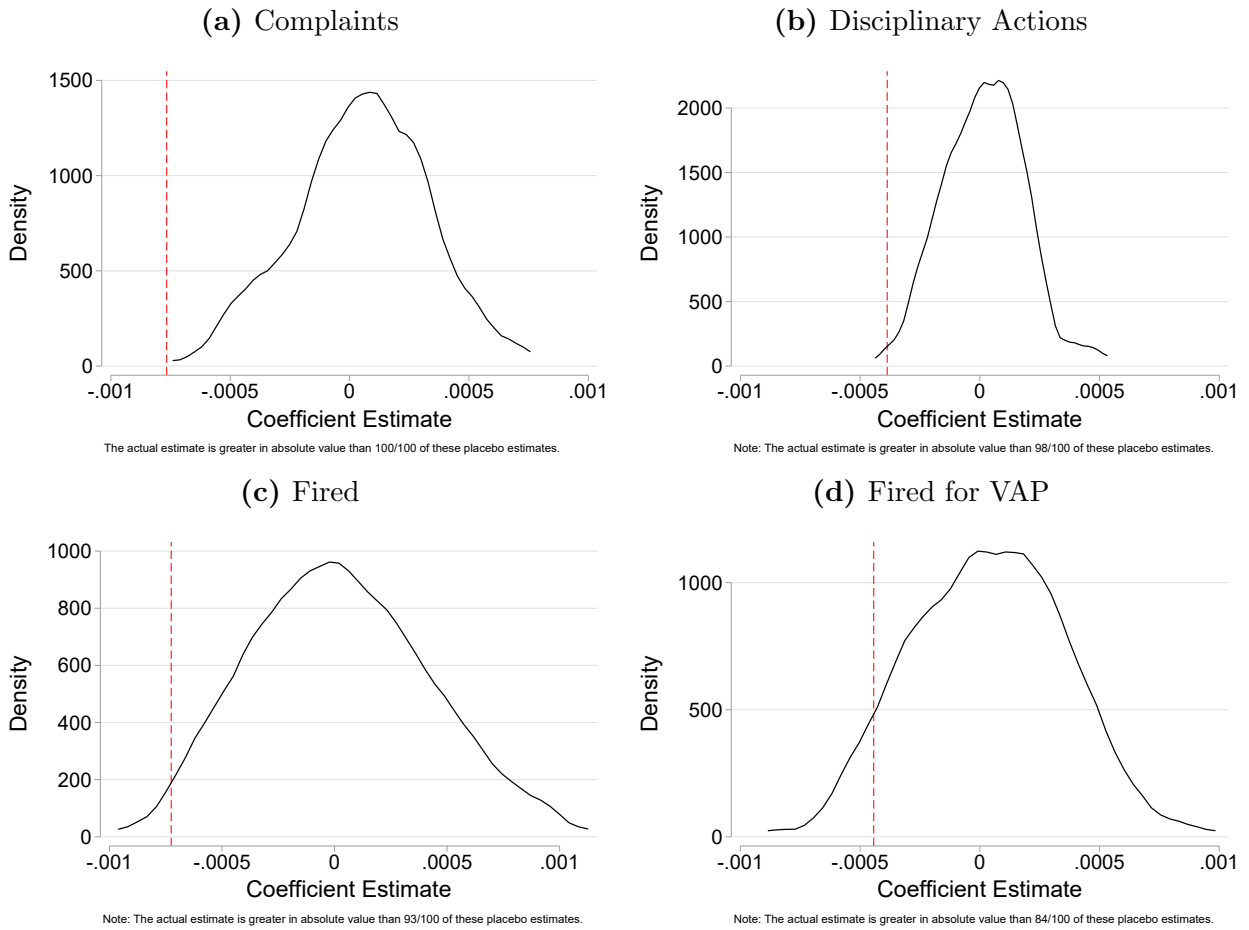


(b) Disciplines



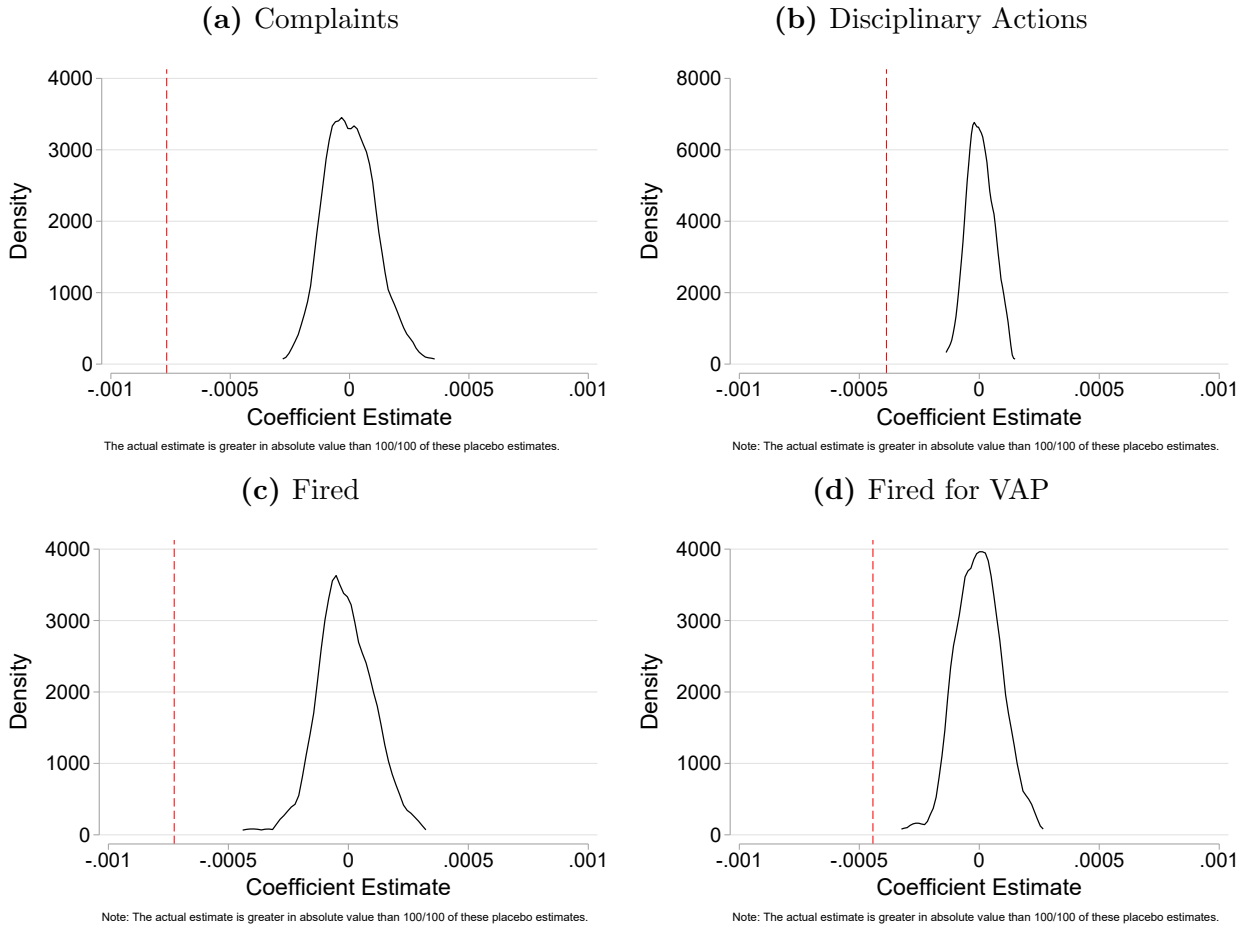
Notes: This figure shows the Kaplan–Meier survival function for police officers until the time they receive a complaint against them (panel (a)) or a discipline against them (panel (b)), measured by the years elapsed since they were hired. The figure presents the survival function of the time until an officer receives a complaint/discipline by whether the officer was hired in an above- or below-median unemployment year.

Figure A8: Placebo Estimates from Randomizing Starting Year for each Cohort



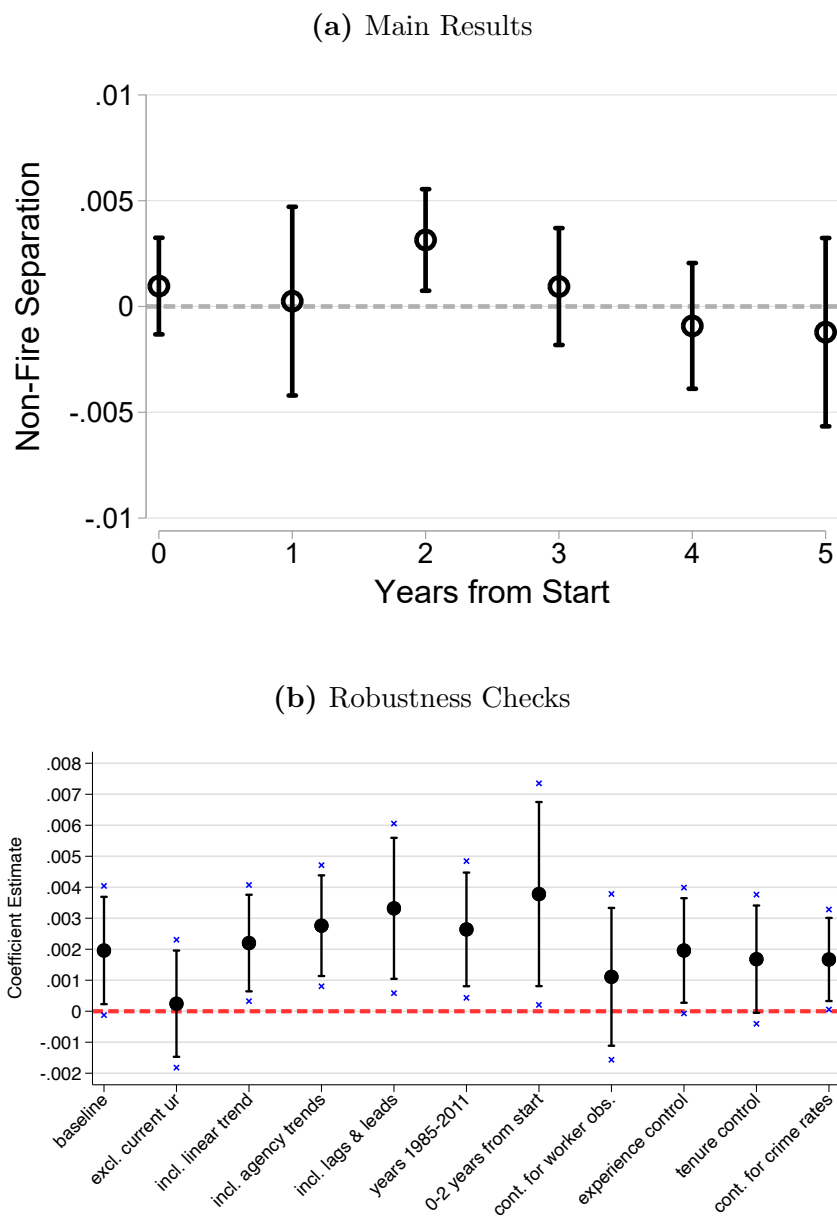
Notes: We randomly assign starting years to entire starting cohorts of police and re-estimate our main results over 100 replications. In this figure, we plot those placebo estimates for our four main outcomes. The actual estimate in each case is marked by the dashed red line. See Table 2 notes for a description of the outcome variables.

Figure A9: Placebo Estimates from Randomizing Starting Year for each Individual



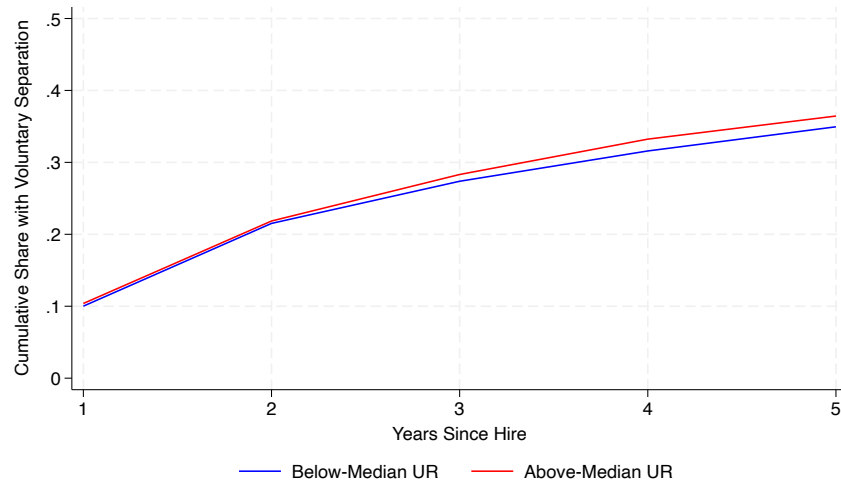
Notes: We randomly assign starting years to individual officers and re-estimate our main results over 100 replications. In this figure, we plot those placebo estimates for our four main outcomes. The actual estimate in each case is marked by the dashed red line. See Table 2 notes for a description of the outcome variables.

Figure A10: Effect of Starting Year Unemployment Rate on Officer Separations



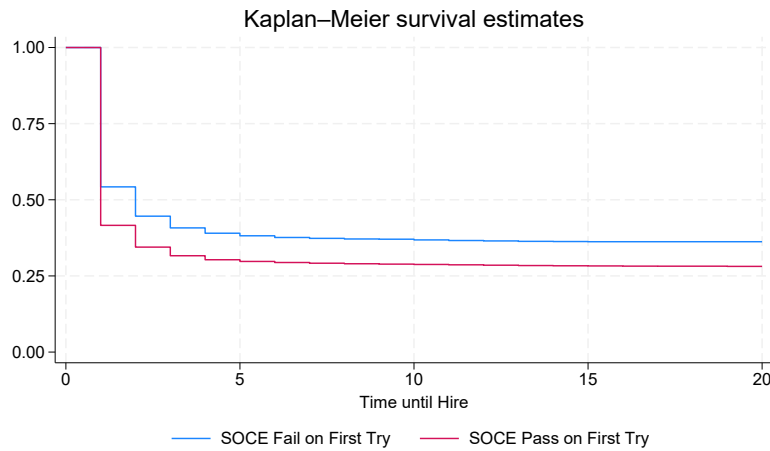
Notes: Panel (a) estimates equation (1) with voluntary separation as the dependent variable separately for each year since start. Panel (b) shows our standard set of robustness checks for this outcome. In both panels, with each coefficient, we plot the 90 percent confidence interval, based on standard errors clustered at the level of officer's starting year.

Figure A11: Voluntary Separation by Initial Unemployment Rate



Notes: This figure plots the cumulative share of officers who voluntarily separate from their agency within five years of being hired for officers hired in years with an above- versus below-median unemployment rate.

Figure A12: Passing the SOCE on the First Try and the Probability of Being Hired



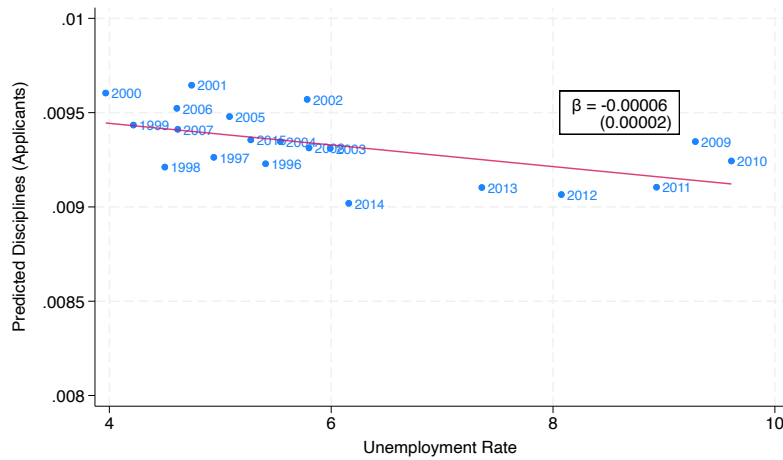
Notes: This figure plots the likelihood of being hired by a police agency in Florida as a function of time since passing the SOCE exam based on non-parametric Kaplan-Meier estimates of the survival function, divided by whether the candidate passed the SOCE exam on the first try or not. The sample only includes officers who eventually passed the SOCE exam (and were thus eligible to be hired), and for whom we observe detailed information on their police training academy information, yielding a sample of 52,195 candidates.

Figure A13: Predicted Misconduct Events Among Applicants and Unemployment Rate

(a) Predicted Complaints



(b) Predicted Disciplines



Notes: This figure examines how applicants' predicted complaints and disciplines (from the machine learning exercise described in Section 4.1.2) vary across the business cycle, as captured by the unemployment rate in 1996-2015. The first panel presents evidence for applicants' predicted complaints, whereas the second panel presents corresponding evidence for predicted disciplines.

Figure A14: Relationship between # of New Hires and Starting Year Unemployment Rate, Statewide Agencies



Notes: This figure plots the number of full-time police hires per year in the Florida Department of Highway Safety and Motor Vehicles and the unemployment rate in each year. This figure suggests that the Florida Department of Highway Safety and Motor Vehicles either does not exhibit a labor demand response to economic conditions or exhibits a labor demand response that is far more muted than other agencies in Florida.